EarthGPT: A Universal Multimodal Large Language Model for Multisensor Image Comprehension in Remote Sensing Domain

Wei Zhang, Miaoxin Cai, *Graduate Student Member, IEEE*, Tong Zhang, *Graduate Student Member, IEEE*, Yin Zhuang, *Member, IEEE*, and Xuerui Mao

***Abstract*— Multimodal large language models (MLLMs) have demonstrated remarkable success in vision and visual-language tasks within the natural image domain. Owing to the significant domain gap between natural and remote sensing (RS) images, the development of MLLMs in the RS domain is still in the infant stage. To fill the gap, a pioneer MLLM named EarthGPT integrating various multisensor RS interpretation tasks uniformly is proposed in this article for universal RS image comprehension. First, a visual-enhanced perception mechanism is constructed to refine and incorporate coarse-scale semantic perception information and fine-scale detailed perception information. Second, a cross-modal mutual comprehension approach is proposed, aiming at enhancing the interplay between visual perception and language comprehension and deepening the comprehension of both visual and language content. Finally, a unified instruction tuning method for multisensor multitasking in the RS domain is proposed to unify a wide range of tasks including scene classification, image captioning, region-level captioning, visual question answering (VQA), visual grounding, and object detection. More importantly, a dataset named MMRS-1M featuring large-scale multisensor multimodal RS instruction-following is constructed, comprising over 1M image–text pairs based on 34 existing diverse RS datasets and including multisensor images such as optical, synthetic aperture radar (SAR), and infrared. The MMRS-1M dataset addresses the drawback of MLLMs on RS expert knowledge and stimulates the development of MLLMs in the RS domain. Extensive experiments are conducted, demonstrating the EarthGPT's superior performance in various RS visual interpretation tasks compared with the other specialist models and MLLMs, proving the effectiveness of the proposed EarthGPT and offering a versatile paradigm for open-set reasoning tasks. Our code and dataset are available at https://github.com/wivizhang/EarthGPT.**

Manuscript received 4 February 2024; revised 3 May 2024; accepted 25 May 2024. Date of publication 4 June 2024; date of current version 18 June 2024. This work was supported by the National Natural Science Foundation of China under Grant 92152109 and in part by the General Program of National Natural Science Foundation of China under Grant 62371048. *(Wei Zhang and Miaoxin Cai are co-first authors.) (Corresponding author: Xuerui Mao.)*

Wei Zhang is with the Advanced Research Institute of Multidisciplinary Sciences and the School of Mechatronical Engineering, Beijing Institute of Technology, Beijing 100081, China (e-mail: w.w.zhanger@gmail.com).

Miaoxin Cai, Tong Zhang, and Yin Zhuang are with the National Key Laboratory of Science and Technology on Space-Born Intelligent Information Processing, Beijing Institute of Technology, Beijing 100081, China (e-mail: 3120220667@bit.edu.cn; bit_zhangtong@163.com; yzhuang@bit.edu.cn).

Xuerui Mao is with the Advanced Research Institute of Multidisciplinary Sciences and the School of Mechatronical Engineering, Beijing Institute of Technology, Beijing 100081, China, and also with the Yangtze Delta Region Academy, Beijing Institute of Technology, Jiaxing 314003, China (e-mail: xmao@bit.edu.cn).

Digital Object Identifier 10.1109/TGRS.2024.3409624

***Index Terms*— Instruction-following, multimodal large language model (MLLM), multisensor, remote sensing (RS).**

I. Introduction

UNIFYING multitask interpretation in the remote sensing (RS) domain is crucial in practical application as real-world scenarios often demand comprehensive analyses to make informed decisions. Although deep learning methods in RS have been successful in RS visual analysis tasks [1], [2], [3], current methods mainly follow a one-task-one-architecture paradigm which limits their ability to handle multisensor RS images, multiple tasks, and generalize to open-set reasoning.

Most recently, the development of multimodal large language models (MLLMs) [4], [5], [6], [7], has shown great success in the natural image domain, as they not only possess robust language interaction capabilities but also exhibit impressive multitask reasoning skills in real-world scenarios [8], compared to smaller and domain-specific models tailored for particular tasks. These MLLMs showed effectiveness in tasks like detailed image description, question answering, and spatial localization guided by language instructions. The remarkable multimodal reasoning ability allows MLLMs to generalize well in new situations and demonstrate powerful zero-shot capabilities across open-set tasks [9]. However, due to the significant domain gap between the nature scene and RS images such as the imaging conditions, viewpoints, different spatial contextual relations, and so on. It is challenging to directly apply nature scene MLLMs to the RS domain.

Although rare MLLM works such as RSGPT [10] and GeoChat [11] have been proposed to explore the MLLMs for the RS field and trying to understand RS images through language context, they still have limitations. Notably, RSGPT possesses image–text caption and visual question-answering (VQA) capabilities but struggles with tasks like classification, detection, and visual grounding. In addition, RSGPT has to tune the model for each task individually, leading to the lack of generalizable ability in the open-set domain. GeoChat has been proposed to explore RS MLLMs and try to solve multiple tasks. Nevertheless, GeoChat is trained on optical RS images and lacks the ability to interpret multiple visual modalities such as synthetic aperture radar (SAR) and infrared modalities. It is clear that the existing RS MLLMs do not establish an

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:46:26 UTC from IEEE Xplore. Restrictions apply.



Fig. 1. EarthGPT is a pioneering model designed to seamlessly unify multisensor and diverse RS intelligent interpretation tasks in a unified framework, guided by user language instructions, and is versatile at performing visual-language dialogs across optical, SAR, and infrared images. EarthGPT's capabilities extend to a wide range of tasks including scene classification, image description, VQA, target description, visual localization, and object detection.

effective unified learning mechanism to achieve multitask, multisensor, and open-set reasoning capabilities uniformly. Therefore, it is indispensable to construct a more universal MLLM tailored to the RS domain.

To leverage the robust generalization and emergent ability of large language models (LLMs), a versatile MLLM called EarthGPT is proposed for multisensor RS image comprehension, unifying various RS interpretation tasks effectively. As illustrated in Fig. 1, EarthGPT can serve as an intelligent assistant capable of efficiently handling a wide range of RS tasks through language interaction.

In EarthGPT, three key techniques are proposed. First, a visual-enhanced perception mechanism is constructed to mix various visual backbones. Specifically, the vision transformer (ViT) [12] and convolutional neural network (CNN) [13] backbones are utilized to capture spatial contextual information and local detailed information, respectively, thereby enhancing visual comprehension. For subsequent alignment with language tokens, all visual features are transformed into 1-D vectors by the liner projection layer. Moreover, the language perception module is employed to encode language instructions into 1-D language embeddings. Second, a cross-modal mutual comprehension approach is proposed. By directly concatenating the visual features with language features to generate multimodal input for LLM, then unfreezing the self-attention and RMSNorm layer of LLM transformer blocks to train on natural common datasets, visual-language alignment is implemented, and the interplay comprehension between visual and language content is deepened. Finally, a unified instruction tuning method for multisensor and multitasking in the RS domain is proposed. Specifically, we continue fine-tuning LLM using the bias-tuning strategy on RS data. MLLM is endowed with the capability of following the language instructions and unifying a wide range of tasks

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:46:26 UTC from IEEE Xplore. Restrictions apply.

including scene classification, image captioning, region-level captioning, VQA, visual grounding, horizontal bounding box (HBB) object detection, and oriented bounding box (OBB) object detection. During the two-step training, the domain transfer learning strategy [3], [14] is adopted, allowing the proposed model to leverage generalized knowledge learned from the broader natural datasets during the cross-modal mutual comprehension phase, and then continue fine-tuning to transfer the general knowledge into the RS domain for better interpreting RS data.

More importantly, a large-scale dataset called MMRS-1M which is a multisensor multimodal RS instruction-following dataset, is created. MMRS-1M comprises over 1M image–text pairs transformed from 34 existing diverse RS datasets and includes multisensor images such as optical, SAR, and infrared. This dataset is tailored to the unique visual modalities and geographical characteristics of the RS domain, effectively mitigating the lack of RS domain expertise of MLLMs. Furthermore, MMRS-1M serves as a catalyst for the advancement of MLLMs in the RS domain.

Extensive experiments are conducted in multiple RS datasets. It shows that the proposed EarthGPT surpasses the state-of-the-art (SOTA) specialist models and MLLMs in most RS tasks. In particular, for the image captioning, VQA, and visual grounding task in the supervised setting, EarthGPT shows a notable improvement in the NWPU-Captions [15], CRSVQA [16], and DIOR-RSVG [17] datasets compared with other specialist models. We also assess generalization ability in the open-set domain for the proposed EarthGPT. For the zero-shot scene classification task, EarthGPT achieves 77.37% and 74.72% accuracy on the CLRS [18] and NaSC-TG2 [19] datasets, respectively, far exceeding other MLLMs. On the MAR20 [20] detection dataset, EarthGPT achieves AP@40% metrics of 90.47% accuracy, outperforming other MLLMs and open-set detection models. In conclusion, experimental results demonstrate EarthGPT's superior performance in a wide range of RS multisensor image comprehension tasks, and robust generalization capability in open-set reasoning tasks.

In summary, the contributions of this article are as follows.

1) To the best of our knowledge, the largest multimodal multisensor RS instruction-following dataset named MMRS-1M is constructed, consisting of over 1M image–text pairs that include optical, SAR, and infrared RS images. This dataset effectively mitigates the lack of RS domain expertise in MLLMs.
2) A pioneer MLLM called EarthGPT is proposed for multisensor image comprehension in the RS domain. EarthGPT effectively establishes a multimodal unified learning framework and unifies a wide range of RS visual and visual-language tasks in the multiturn dialog format.
3) The visual-enhanced perception mechanism is designed, to capture coarse-scale spatial contextual information and fine-scale local information signal to facilitate image interpretation. Furthermore, the cross-modal mutual comprehension approach is developed for visual-language alignment and converting the language-only LLM into an MLLM, which is endowed with general visual understanding ability.
4) The unified instruction tuning method is proposed, enhancing EarthGPT's ability to follow language instructions for various RS downstream tasks, and adapt the natural domain generalized knowledge to the RS domain for achieving robust open-set reasoning ability.
5) Extensive experiments demonstrate the EarthGPT's superior performance in multisensor RS visual interpretation tasks compared with the other specialist models and MLLMs in scene classification, image captioning, region-level captioning, VQA, visual grounding, and object detection. EarthGPT therefore represents a significant advance of MLLMs in the field of RS and contributes a versatile paradigm for RS visual-language mutual comprehension.

II. Related Work

A. Multimodal LLMs

The continuous emergence of LLMs has fueled rapid advancements in natural language processing (NLP), showcasing extraordinary capabilities in language modeling across diverse contexts. Pretraining on massive text corpora, fine-tuning on specific domains, and the continual expansion of model parameters have enabled LLMs to achieve new milestones in various benchmark tests. Notably, OpenAI's ChatGPT [21] addresses users' needs across different scenarios through human–machine dialogs and text generations. The LLaMA-guided fine-tuning models [22], [23], [24] have gained favor among LLM researchers. In addition, GPT-4-LLM [25] demonstrates significant improvements when provided with high-quality instruction-following datasets. As a fundamental component of general intelligence, multimodal perception is a crucial step toward achieving universal models. Researchers [4], [5], [6], [7] are currently devoted to incorporating multimodal data beyond text into LLM to support specific tasks across various modalities. Models like VisualGPT [4], BLIP [5], Flamingo [7], and Kosmos-1 [26] showcase strong multimodal reasoning potential after aligning LLMs with image modalities. To achieve modality alignment, many works, such as MiniGPT-4 [9], LLAMA-Adapter V2 [27], mPLUG-Owl [28], employed projection layers, zero-shot attention mechanism, and intermediate networks to fuse LLaMA and visual modality. With the continuous progress of MLLMs, additional modalities such as audio, video [29], [30], and 3-D point clouds [31], [32], [33] are being integrated and aligned. The extraordinary potential of MLLMs is also evident in fields such as LIDAR [34] and robotics [35]. Nevertheless, the MLLMs mentioned above are tailored for natural scenes, missing the capacity to capture the distinctive contextual complexities of the RS domain.

Recently, there has been an emergence of MLLMs for RS. For instance, RSGPT [10] fine-tunes Instruct-BLIP [36] on a high-quality image–text pair dataset, showing good image–text caption and VQA capabilities. However, RSGPT faces difficulties in performing tasks such as classification and detection. In addition, GeoChat [11] has been proposed to

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:46:26 UTC from IEEE Xplore. Restrictions apply.

explore RS MLLMs and to solve multiple tasks on optical RS images. However, GeoChat is incapable of understanding multisensor images including SAR and infrared modalities. To address the limitations of existing MLLMs, this article focuses on developing a more universal RS MLLM, which seamlessly integrates multiple RS tasks and visual modalities from multisensors.

B. RS Interpretation Tasks and Specialists

RS image interpretation is crucial for monitoring and analyzing the Earth's surface and environment, with widespread applications in military defense, geological surveying, and urban planning [1], [2], [3]. Image interpretation tasks include scene classification, image captioning, VQA, visual grounding (VG), and object detection. Specifically, the scene classification task aims at labeling RS images with predefined categories based on their contents. Many specialist models such as SeCo [37], CSDS [38], MGSNet [39], T-CNN [40], and PSGAN [41], have made progress on scene classification. For image captioning, it aims to generate descriptive sentences that accurately describe the content of an RS image. Representative work under this task involves CSMLF [42], FC-Att [43], SM-Att [43], and MLCA-Net [15]. Subsequently, the VQA task in its ideal form is to let models achieve reasoning in the joint space of vision and language, requiring an understanding of both the image content and the question. The typical specialist models on VQA includes Qonly [44], RSVQA [45], SAN [46], and MQVQA [16]. Moreover, the VG task aims to localize the object referred by the query expression in RS images, the models such as ZSGNet [47], FAOA [48], LBYL-Net [49], and TransVG [50], contributing to this topic. Finally, object detection aims at locating and classifying objects within an RS image by predicting bounding boxes and their categories. The detection specialists comprises S2A-Net [51], CFA [52], Oriented R-CNN [53], R3Det [54], and Sasm [55]. In conclusion, those specialists contribute to advancing RS image interpretation. However, they are designed to deal with a single modality or single task.

Apparently, those specialists mentioned above cannot address multiple visual tasks within an architecture, exhibiting poor scalability, which significantly impacts the practical application. Therefore, in this article, our goal is to develop an MLLM for the RS domain to unify a wide range of RS visual comprehension tasks.

C. RS Datasets

RS datasets are the core and foundation of RS intelligent interpretation models. Currently, RS datasets mainly focus on classification, detection, segmentation, image captioning, and VQA tasks. For classification, commonly used datasets include AID [56], EuroSAT [57], NWPU-RESISC45 [58], UCMerced LandUse [59], and WHU-RS19 [60]. For detection, optical datasets like DIOR [61], DOTA [62], FAIR1M [63], HRRSD [64], NWPUVHR10 [65], and SAR datasets like AIR-SARShip-2.0 [66], HRSID [67], and SSDD [68] are used to train expert models. Segmentation datasets typically include Vaihingen, Potsdam, iSAID [69], and LoveDA [70]. Image captioning datasets include Sydney-Captions [71], RSICD [72], NWPU-Captions [15], RSITMD [73], and UCM-Captions [71] are used for generating image descriptions or image–text retrieval. VQA datasets mainly consist of Floodnet [74], RSVQA-LR [45], RSVQA-HR [45], RSIVQA [75], and CRSVQA [16]. The datasets mentioned above primarily emphasize a single visual modality within an individual task, leading to models with limited generalization capabilities and restricted to specific tasks. MLLMs represent a potential solution to tackle the challenge. However, in the current landscape of the RS domain, there is a scarcity of high-quality image–text instruction-following data. This scarcity poses a significant obstacle when it comes to training multimodal intelligent dialog assistants that can achieve a high level of alignment between visual and language, unify various tasks into a framework, and integrate different multisensor visual modalities effectively.

To address this challenge, we have constructed a dataset called MMRS-1M containing more than 1M image–text pairs, covering tasks such as classification, detection, image captioning, VQA, and visual grounding. MMRS-1M encompasses three visual modalities such as optical, infrared, and SAR. MMRS-1M is designed to promote the continuous development of MLLMs in the RS field.

III. Methodology

This section introduces the proposed EarthGPT framework. We first overview the overall model architecture (see Section III-A). Subsequently, four fundamental techniques in EarthGPT are elaborated, involving the visual-enhanced perception mechanism (see Section III-B), the language perception (see Section III-C), the cross-modal mutual comprehension approach (see Section III-D), and the unified instruction tuning method for RS (see Section III-E).

A. Overview

Our goal is to break down the differences between natural images and RS images that lead to the obstacles to the application of MLLMs in the RS domain and propose a universal MLLM specifically customized for RS. To this end, a unified MLLM called EarthGPT is proposed for RS imagery comprehension. The overall framework of EarthGPT is shown in Fig. 2(a). Note that four pivotal techniques are developed. First, to tackle the challenge that RS images commonly contain various disturbances that compromise clarity, the visual-enhanced perception mechanism is proposed, as shown in step 1. Specifically, ViT and CNN backbones are designed to refine global and detailed visual features through multilayer and multiscale visual perception. Then, twofold visual tokens are transformed by the linear projection layer to ensure dimension alignment and then concatenated together. The hybrid visual tokens contain both neighboring dependencies and long-range visual interactions, offering subtle local insights and extensive regional correlations. Meanwhile, the language instruction is converted into language tokens by the language perception module. Subsequently, the cross-modal mutual comprehension approach is developed in step 2. This

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:46:26 UTC from IEEE Xplore. Restrictions apply.

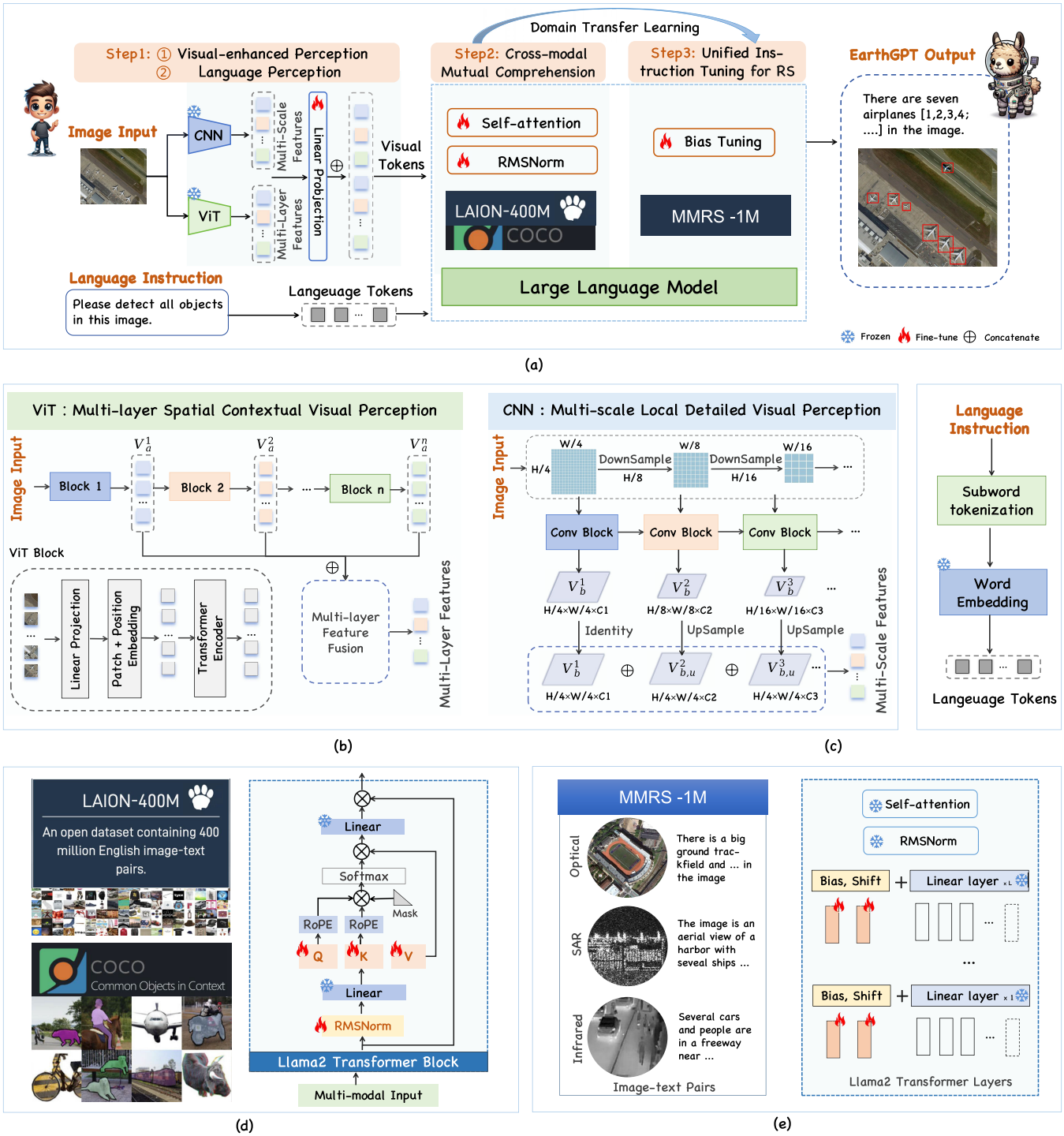


Fig. 2. (a) Overall model architecture of EarthGPT. (b) Detailed illustration of visual-enhanced perception mechanism. (c) Detailed illustration of language perception mechanism. (d) Detailed illustration of the cross-modal mutual comprehension approach. (e) Detailed illustration of the unified instruction tuning method for RS.

part is to equip the LLM with fundamental visual comprehension capabilities and achieve visual-language alignment through tuning the LLM on nature scene datasets. Particularly, the visual and language tokens are merged as the multimodal input for the unfrozen LLM tuning to enhance the interaction between the diverse visual and linguistic contexts. Finally, to overcome the limitations of existing RS specialist models, which are unable to handle multisensor data and multitasks within one framework, a unified instruction tuning method for RS is developed in step 3. Concretely, the bias tuning for linear layers in LLM is performed on the newly constructed MMRS-1M dataset, achieving the universal visual interpretation in the RS domain. In the two-step training, the domain transfer learning strategy is adopted to transfer the general knowledge into the RS domain to facilitate RS data comprehension. We will introduce each technique as follows in detail.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:46:26 UTC from IEEE Xplore. Restrictions apply.

B. Visual-Enhanced Perception

The core of the designed visual-enhanced perception mechanism is to utilize the strengths of diverse image encoding models to enhance and refine multigranularity critical visual information while simultaneously mitigating various disturbances in RS images. Specifically, we utilize ViT [12] to capture the spatial contextual information at different levels for images and employ CNN [13] to extract the local information present across multiple scales. After mixed encoding, the comprehensive extraction of visual information is achieved, preparing for alignment with LLM. The visual-enhanced perception mechanism consists of two modules as shown below.

1) Multilayer Spatial Contextual Visual Perception: We adopt the ViT as the image encoder to extract multilayer features, as shown in the left of Fig. 2(b). The ViT model splits the input image into patch sequences, allowing models to learn image structures independently. Then, the flattened patches are linearly embedded and added with position embeddings, and the resulting sequence of vectors is fed to a standard transformer encoder. Subsequently, multilayer visual intermediate features $\{V_a^i\}_{i=1}^n$ are extracted from different layers to capture long-range contextual spatial information, where n represents the number of layers. The original images I are transformed into visual representations V_a^1. Then, all the extracted visual features are concatenated along the channel dimension. Such visual feature fusion from different layers is beneficial to capture subtle differences in RS images and provides a more comprehensive understanding of images. The entire feature extraction process can be given as

$$V_a^1 = \text{ViTBlock}_1(I) \tag{1}$$

$$V_a^i = \text{ViTBlock}_i\left(V_a^{i-1}\right), \quad i = 2, \ldots, n \tag{2}$$

$$V_a = \text{Concat}\left[V_a^1, V_a^2, \ldots, V_a^n\right]. \tag{3}$$

In this way, the extracted features both involve spatial-aware information from early layers and semantic indicative from later layers.

2) Multiscale Local Detailed Visual Perception: To integrate multiscale local details into visual representations, the CNN backbone is designed as the image encoder. Compared to ViT, CNN excels in extracting localized features, such as edges and textures, through its inherent spatial hierarchies and local receptive fields. The extracted multiscale visual features are denoted as $\{V_b^i\}_{i=1}^m$, where m represents the image scale numbers. As illustrated at the right of Fig. 2(b), the input of CNN includes multiple spatial resolutions of $(H/4) \times (W/4)$, $(H/8) \times (W/8)$, $(H/16) \times (W/16)$, and so on, where $H \times W$ is the original input image resolution. The convolution blocks convert the inputs into token embeddings $V_b^1 \in \mathbb{R}^{(H/4)\times(W/4)\times C_1}$, $V_b^2 \in \mathbb{R}^{(H/8)\times(W/8)\times C_2}$, $V_b^3 \in \mathbb{R}^{(H/16)\times(W/16)\times C_3}$, and so on. Then, the visual features V_b^2, V_b^3, and so on, are upsampled to match the feature resolution of V_b^1, respectively. Hence, all visual features are transformed into the same resolution, denoted as $V_b^1, V_{b,u}^2, V_{b,u}^3, \ldots, V_{b,u}^m$. Afterward, they are concatenated together along the channel dimension. The feature extraction process can be written as

$$V_b^1 = \text{ConvBlock}_1(I_1) \tag{4}$$

$$V_b^i = \text{ConvBlock}_i\left(V_b^{i-1}\right), \quad i = 2, \ldots, m \tag{5}$$

$$V_{b,u}^i = \text{Upsample}_i\left(V_b^i\right), \quad i = 2, \ldots, m \tag{6}$$

$$V_b = \text{Concat}\left[V_b^1, V_{b,u}^2, V_{b,u}^3, \ldots, V_{b,u}^m\right] \tag{7}$$

where I_1 represents the first scale input. The multiscale fused visual features encompass information of multigranularity, enabling the capture of both broad semantics and intricate information.

As shown in Fig. 2(a), after extracting visual features using ViT and CNN backbones, a learnable projection layer is employed to flatten visual features into 1-D vectors for facilitating subsequent alignment with language tokens. Then, multilayer and multiscale features are concatenated channel-wisely. The new visual tokens are denoted as V_p. This process is formulated as

$$V_p = \text{Concat}(\text{Projection}(V_a), \text{Projection}(V_b)). \tag{8}$$

The integration of enhanced visual perception information improves the accuracy and efficiency of the image interpretation.

C. Language Perception

Visual tokens are obtained based on the aforementioned visual-enhanced perception mechanism. Meanwhile, language perception is also proposed. The core of the language perception mechanism is to transform language instructions into language embeddings, preparing for the next step and injecting them as textual information into LLM.

Following most NLP models [22], [76], [77], the common language tokenizer is employed to encode language instructions. As shown in Fig. 2(c), we adopt the subword tokenization [78] method to divide instructions into subwords. This method is beneficial for understanding the structure of the text and preparing for further processing like embedding. After tokenization, the next step is to convert these subwords into embeddings. The Word2Vec [79] technique is employed to map each subword to a 1-D vector embedding, denoted as L_p. The language embeddings serve as a bridge between the textual instructions data and the LLMs, enabling them to process and analyze language data inputted by the user.

D. Cross-Modal Mutual Comprehension

The key point of the cross-modal mutual comprehension approach lies in visual and language perception information fusion and delivering the multimodal input into the LLM for alignment and interaction training.

To form multimodal LLM input tokens $\mathcal{X}$, the overall visual embeddings V_p are concatenated with the language instruction embeddings L_p. The process is expressed as

$$\mathcal{X} = \text{Concat}\left[\overbrace{V_p^1, V_p^2, \ldots, V_p^{N_v}}^{\text{visual tokens } V_p}, \underbrace{L_p^1, L_p^2, \ldots, L_p^{N_l}}_{\text{language tokens } L_p}\right] \tag{9}$$

where N_v represents the token length of visual features, N_l represents the token length of language features, $(V_p^1, V_p^2, \ldots, V_p^{N_v})$ are the mixed visual tokens from V_p, and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:46:26 UTC from IEEE Xplore. Restrictions apply.

$(L_p^1, L_p^2, \ldots, L_p^{N_l})$ are the language instruction tokens from L_p. Then, multimodal input $\mathcal{X}$ is fed into LLM for fusion and integration.

In current MLLMs [4], [5], [6], [7], [9], visual-language understanding is achieved through training on a frozen LLM, designed to avoid expensive full-parameter fine-tuning. However, freezing all the LLM's weights significantly limits its potential for comprehensive cross-modal learning. To cope with the challenge of the knowledge gap from different modalities and realize multimodal mutual comprehension, an unfrozen visual-language alignment strategy is designed. Specifically, the self-attention and normalization layers are unfrozen for training on the most common domain data (LAION-400M [80] and COCO Caption [81]). This strategy aids the LLM in thoroughly understanding multimodal representations and simultaneously preserves the inherent language response capabilities of LLM through the partly frozen modules.

In our method, Llama [22] with powerful language understanding capability is adopted as the initial LLM. Llama 2 is composed of L transformer blocks. As shown in Fig. 2(d), the transformer block consists of self-attention, RMSNorm, FFN layers, rotary position embedding (RoPE), softmax, and mask module. We only unfreeze part parameters for training, without modifying the internal structure of Llama 2. Specifically, the RoPE module is to encode positional information. The mask module ensures that when LLM generates content of the current position, only content that has already been generated, preventing the model from "seeing" the text that has not yet been generated. The softmax function converts the probability distribution at each position into the probability of corresponding tokens. The RoPE, mask, and softmax modules do not participate in parameter updates. In addition, the linear layers are frozen in this training stage. Most importantly, Llama 2's key module self-attention head is composed of key $\mathbf{K}$, query $\mathbf{Q}$, and the value $\mathbf{V}$, as shown in Fig. 2(d). The $\mathbf{K}$, $\mathbf{Q}$, and $\mathbf{V}$ are implemented via linear layers. The detailed implementation is given as follows:

$$\mathbf{Q}(\mathcal{X}) = \mathbf{W}_q\ \mathcal{X} + \mathbf{b}_q \tag{10}$$

$$\mathbf{K}(\mathcal{X}) = \mathbf{W}_k\ \mathcal{X} + \mathbf{b}_k \tag{11}$$

$$\mathbf{V}(\mathcal{X}) = \mathbf{W}_v\ \mathcal{X} + \mathbf{b}_v. \tag{12}$$

The parameters $\mathbf{W}_q$, $\mathbf{W}_k$, $\mathbf{W}_v$, $\mathbf{b}_q$, $\mathbf{b}_k$, and $\mathbf{b}_v$ are updated during the training. The RMSNorm component of the transformer is also unfrozen, and the scale operation γ is set as a trainable parameter. The RMSNorm can be expressed as

$$y = \frac{x}{\sqrt{\text{Mean}\left(x^2\right) + \varepsilon}} * \gamma \tag{13}$$

where

$$\text{Mean}\left(x^2\right) = \frac{1}{N}\sum_{i=1}^{N} x_i^2. \tag{14}$$

During the training process, the adopted cross-entropy loss function is described as $\mathcal{L}$, the multimodal input sequence length is denoted as N_{vl}, and the parameters of EarthGPT are represented as θ. The cross-entropy loss function $\mathcal{L}$ can be formulated as

$$\mathcal{L} = -\sum_{i=1}^{N_{\text{vl}}} \log P(w_i | (w_1, w_2, \ldots, w_{i-1}; \theta)) \tag{15}$$

where w_i denotes the ith word. After visual-language alignment, the language-only LLM is converted into an MLLM, which can generate the visual interpretation response based on the integrated multimodal information.

E. Unified Instruction Tuning for RS

After the cross-modal mutual comprehension training, the proposed EarthGPT has basic multimodal reasoning and dialog capabilities in the natural domain but struggles to follow instructions and perform inference for RS downstream tasks in the aforementioned stage. To enhance EarthGPT's ability to follow instructions for various downstream tasks and expand its applicability from the natural domain to the RS one, we develop an extensive and unified instruction-following dataset called MMRS-1M. In the MMRS-1M dataset, we have standardized all downstream tasks into the format of VQA instructions. By utilizing MMRS-1M for tuning, the model excels in accurately understanding and executing various instructions in the RS domain.

Technically, to preserve the visual captioning capability obtained in the previous stage and simultaneously enhance compliance with task instructions, we freeze all the weights from the cross-modal mutual comprehension phase and introduce new learnable parameters into the Llama 2 model. Specifically, inspired by LLaMA-Adapter V2 [27], the linear layer has been modified by introducing two learnable parameters including a bias β and a shift α into linear layers, as shown in Fig. 2(e). Given a linear layer $y(x) = \mathbf{W}x$, it can be transformed by incorporating the bias factor and shift factor. The process can be expressed as

$$y(x) = \alpha \cdot (\mathbf{W}x + \beta) \tag{16}$$

where

$$\alpha \sim \mathcal{N}\left(\mu, \sigma^2\right), \quad \beta = \text{Init}(0). \tag{17}$$

The bias β is initialized with zeros, and the shift α is initialized with a random Gaussian, ensuring training stability and effectiveness. Analogously, the loss function in this stage is given as (15).

Leveraging the advanced reasoning abilities of LLM and our abundant MMRS-1M dataset, EarthGPT is equipped with versatile skills for multisensor visual comprehension in the RS domain guided by the language instructions and shows the potential for real-world practical applications.

IV. Dataset Construction

Currently, lacking domain-specific datasets for RS hampers the effective application of MLLMs in the intelligent interpretation of geographic information and open-set dialogs. Therefore, a diverse and comprehensive instruction-following dataset for image–text conversation in RS imagery is indispensable. To address this limitation, a large quantity of existing

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:46:26 UTC from IEEE Xplore. Restrictions apply.

TABLE I
DETAILS ON THE TRAINING SAMPLES USED FOR THE MMRS-1M

Task	Data	Size	Type
Image Captioning	RSICD	24,333	optical
	UCM-Captions	9,986	optical
	RSITMD	20,096	optical
	Sydney-Captions	2,837	optical
	NWPU-Captions	141,631	optical
VQA	FloodNet	4,511	optical
	RSVQA-LR	67,228	optical
	RSIVQA	68,625	optical
	CRSVQA	900	optical
Classification	NWPU-RESISC45	6,300	optical
	EuroSAT	5,400	optical
	UCMerced-Landuse	420	optical
	WHU-RS19	196	optical
	RSSCN7	560	optical
	DSCR	11,951	optical
	FGSCR-42	3,878	optical
Detection	DOTA	163,486	optical
	DIOR	58,200	optical
	FAR1M	40,466	optical
	NWPUVHR10	3,190	optical
	HRRSD	23,044	optical
	RSOD	747	optical
	UCAS-AOD	1,510	optical
	VisDrone	186,810	optical
	AIR-SARShip-2.0	1,433	SAR
	SSDD	1,856	SAR
	HRISD	7,265	SAR
	HIT-UAV	10,608	infrared
	Sea-shipping	16,023	infrared
	Infrared-security	17,234	infrared
	Aerial-mancar	33,051	infrared
	Double-light-vehicle	7,922	infrared
	Oceanic ship	2,505	infrared
Visual Grounding	DIOR-RSVG	30,820	optical
Region-level Captioning	DIOR-RSVG	30,820	optical

RS datasets are carefully cleaned and transformed, and a new instruction-following dataset called MMRS-1M is created covering five tasks (e.g., classification, detection, image caption, VQA, and visual grounding) and three visual modalities (e.g., optical, SAR, and infrared) is created. EarthGPT is fine-tuned on MMRS-1M to align visual and language modalities and achieve excellent coarse-grained conversation and fine-grained localization capabilities. The multimodal and multiturn conversation dataset construction process is described in detail as follows.

A. Multiturn Conversation Data

To leverage existing visual and textual data to develop the multimodal multiturn dialog ability of EarthGPT, a unified data format of image-instruction-answer is transformed. Specifically, for each image I, we transform the existing RS multitask datasets (see Table I) into a format where one image is followed by several instruction-answer pairs. Then, we concat all instruction-answer pairs to generate a long conversation sequence, which is represented as $(I^1_{\text{Instruct}}, I^1_A, I^2_{\text{Instruct}}, I^2_A, \ldots, I^T_{\text{Instruct}}, I^T_A,)$, where T represents the total number of turns, Instruct denotes the language instruction, and A represents the answer. In Sections IV-B and IV-C, we will delve into the detailed processes involved in constructing various visual task-oriented multiturn conversation data, including coarse-grained and fine-grained conversation scenarios.

B. Coarse-Grained Conversation Scenarios

To endow EarthGPT with image-level coarse-grained question-answer capabilities, ten classification datasets, five image captioning datasets, and four VQA datasets are collected for instruction fine-tuning.

1) Category-to-Instruction: The classification datasets include the scene classification datasets AID [56], EuroSAT [57], NWPU-RESISC45 [58], UCMerced-LandUse [59], WHU-RS19 [60], and RSSCN7 [82], as well as the ship classification datasets FGSCR-42 [83] and DSCR [84]. For the format conversion of the classification datasets, category-to-instruction is adopted. Specifically, the category of each image is first extracted, and the template "What is the category of this RS image? Answer the question using a single word or phrase. Reference categories include category 1, . . . , and category n" is used to inquire about the image. The GPT model provides a response that includes the name of the category to which the image belongs.

2) Caption-to-Instruction: First, for image caption datasets, the existing datasets including Syndney-Captions [71], RSICD [72], NWPU-Captions [15], RSITMD [73], and UCM-Captions [71] are cleaned, removing duplicate captions for the same image. Then, the instruction "Please provide a one-sentence caption for the provided RS image in detail." is provided and the EarthGPT model is tasked with describing the image. The number of captions per image determines the number of conversation rounds. Putting multiple captions for each image into a multiturn dialog can significantly reduce computation costs while ensuring diverse descriptions of images without compromising information leakage. This conversion process is denoted as caption-to-instruction.

3) VQA-to-Instruction: For VQA datasets including Floodnet [74], RSVQA-LR [45], RSIVQA [75], and CRSVQA [16], the instruction "Answer the question using a single word or phrase" is added after the original question to control the output format of the answer. The number of question-answer pairs per image determines the number of dialog rounds in the conversation. This multiround dialog allows EarthGPT to extract and interpret sufficient information from the images.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:46:26 UTC from IEEE Xplore. Restrictions apply.

C. Fine-Grained Conversation Scenarios

In order to enable the RS intelligent assistant to perform target region localization and object detection under various imaging conditions, a collection of optical, SAR, and infrared object detection datasets has been assembled. The optical datasets include DIOR [61], DOTA [62], FAIR1M [63], HRRSD [64], NWPUVHR10 [65], RSOD [85], UCAS-AOD [86], and VisDrone [87]. Furthermore, SAR object detection datasets like AIR-SARShip-2.0 [66], HRISD [67], and SSDD [68] are included, along with infrared object detection datasets for HIT-UAV [88], sea-shipping [89], infrared-security [90], aerial-mancar [91], double-light-vehicle [92], and oceanic ship [93]. In addition, the DIOR-RSVG [17] is collected for visual grounding and region-level caption.

1) Detection-to-Instruction: Object detection is divided into HBB and OBB formats. The HBB format defines a bounding box as $[x_{\min}, y_{\min}, x_{\max}, y_{\max}]$. Here, $(x_{\min}, y_{\min})$ and $(x_{\max}, y_{\max})$ represent the corner points of the bounding box closest and farthest from the coordinate origin, respectively. On the other hand, the OBB format is defined as $[x_1, y_1, x_2, y_2, x_3, y_3, x_4, y_4]$. In this format, (x_1, y_1) represents the corner point of the bounding box closest to the coordinate origin, and the remaining three points are sorted in ascending order based on their angles with respect to (x_1, y_1). In addition, for both HBB and OBB detection, the coordinates of the bounding box are normalized. During the dataset conversion process, specific instructions are used to guide the model in predicting either HBB or OBB. This process is referred to as "detection-to-instruction." The instruction "Detect all objects shown in the RS image and describe using HBBs" is used for HBB detection, while "Detect all objects shown in the RS image and describe using OBBs" is used for OBB detection. The model then outputs the coordinates and categories of all objects in the image. Furthermore, to achieve referring detection, an additional multiround conversation session is incorporated to detect objects of each category.

2) VG-to-Instruction: For the visual grounding dataset including DIOR-RSVG, two types of conversation formats are constructed via VG-to-instruction. Specifically, in the first format, users describe the region of interest and provide instructions for MLLM to locate the corresponding target. The model then outputs the coordinates of the identified target. The second format involves the user inputting instructions to describe the target region based on its spatial location coordinates, and MLLM generates a description of the corresponding target region. The former format bestows EarthGPT with the ability to perform visual localization, while the latter is employed to accomplish the region caption task.

The aforementioned is a detailed description of the MMRS-1M dataset construction process. In addition, for the datasets mentioned above, the training set and validation are processed through data transformation and the construction of MMRS-1M after image cutting and dataset splitting. The details on the training samples can be found in Table I. The construction flow, as depicted in Fig. 3, provides a more concrete and intuitive understanding of the dataset construction process.

In summary, we clean and convert 34 various existing RS datasets into a uniform visual instruction-following format, covering optical, SAR, and infrared visual modalities, facilitating the application of MLLMs in the RS domain.

V. Experiments and Analysis

In this section, we conduct extensive experiments to validate the performance of our proposed EarthGPT. In Section V-A, we describe the implementation details. Subsequently, we demonstrate EarthGPT's powerful ability in various scenarios such as classification, image captioning, VQA, visual grounding, and object detection, compared to other MLMs, specialist models, and open-set models.

A. Implementation Details

In the visual-enhanced perception part, the DINOv2 ViT-L/14 [94] is adopted as the ViT encoder, and the frozen CLIP ConvNeXt-L [95] is utilized as the CNN encoder. The two visual encoders are kept frozen with their originally pretrained weights throughout the training. The linear projection layer is initialized randomly and fine-tuned in this stage. In the language perception part, the word embedding module also remains frozen during the training. Subsequently, in the cross-modal mutual comprehension stage, the objective is to convert language-only LLM Llama 2 [24] into an MLLM. We train EarthGPT on LAION-400M [80], COCO Caption [81] datasets, which primarily focus on natural scene image captioning, to develop basic multimodal alignment from scratch. Furthermore, in the unified multitask tuning phase, the goal is to equip MLLM with the versatility needed for diverse RS downstream tasks. We train EarthGPT on the newly constructed dataset MMRS-1M characterizing multiturn conversation instruction-following. Based on the multiturn image–text sequence, we perform two-step training of the Llama 2 for the prediction of the next token, using the auto-regressive training objective function, shown in (15).

During the training process, we train an off-the-shelf 13B language model Llama 2. We use the AdamW optimizer [96] $(\beta 1, \beta 2) = (0.9, 0.95)$, with maximum learning rate of 2×10^{-5}, minimum learning rate of 0. The cross-modal mutual comprehension stage takes about 258 h with 16 A100 GPUs, and the unified instruction tuning stage takes about 96 h with 16 A100 GPUs. The maximum training sequence length is set to 3072.

B. Scene Classification

To evaluate the classification performance, supervised and zero-shot classification assessments are conducted. In the first evaluation, the test sets of NWPU-RESISC45 are utilized. The NWPU-RESISC45 dataset is a large-scale publicly available dataset for RS image scene classification, published by Northwestern Polytechnical University. It is comprised of 45 scene categories, with each category containing 700 RS images. Each image has a size of 256 × 256 pixels; 80% of the dataset is used as the test set to evaluate and compare EarthGPT with other specialist models.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:46:26 UTC from IEEE Xplore. Restrictions apply.

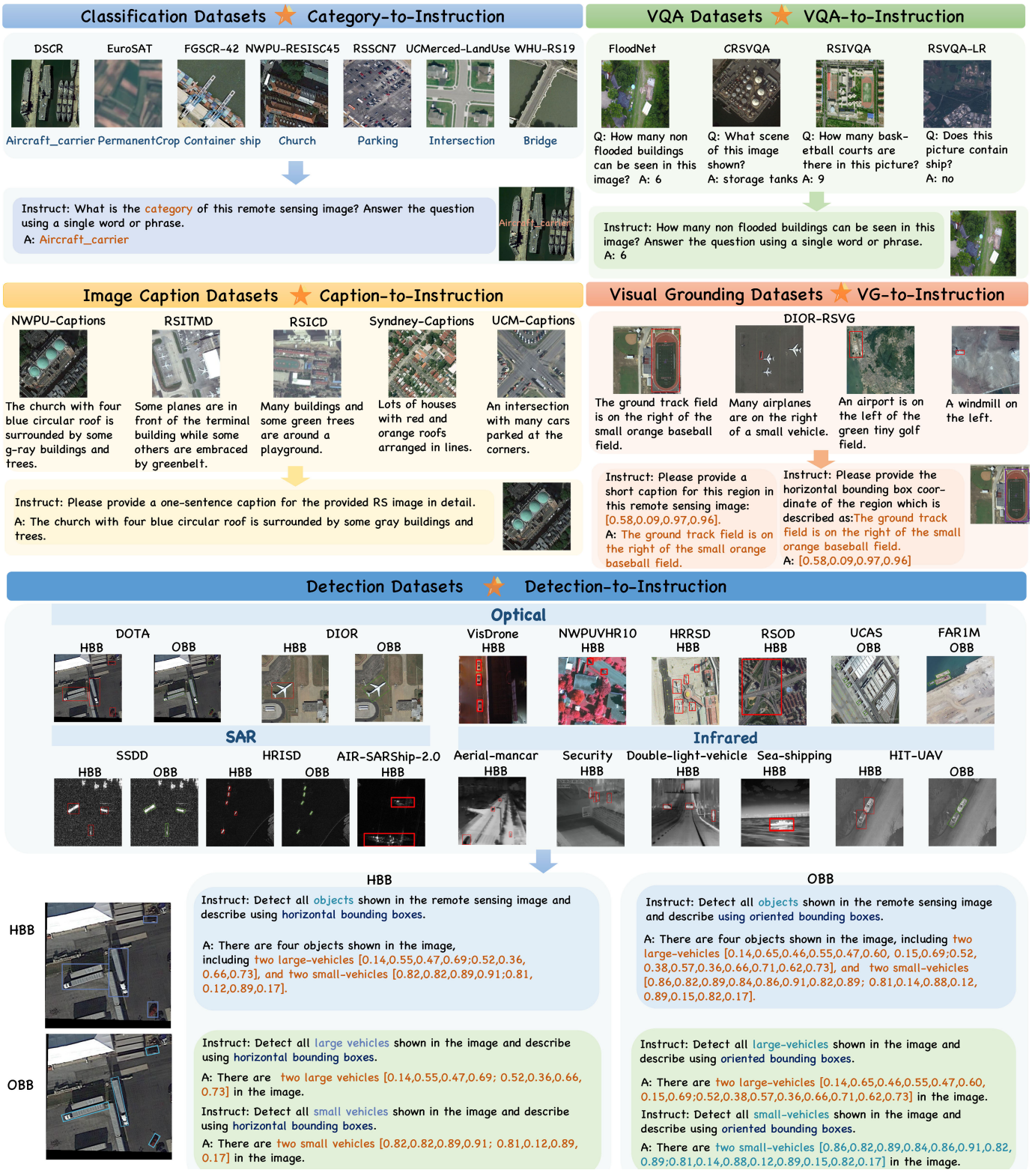


Fig. 3. Construction process of MMRS-1M dataset. MMRS-1M contains three visual modalities from multisensor (e.g., optical, SAR, and infrared) and five RS vision task data (e.g., classification, detection, image caption, VQA, and visual grounding).

For zero-shot classification, all the images from the CLRS [18] and NaSC-TG2 [19] datasets are employed as the zero-shot testing set. The CLRS dataset includes 15 000 RS images with 25 land-use types such as airports and beaches. These images were extracted from Google Earth, Bing Maps, Google Maps, and Tianditu, and were released by Central South University in 2020. Each image has a size of 256 × 256 × 3, and the resolutions range from 0.26 to 8.85 m. The NaSC-TG2 dataset consists of 20 000 RS images of 10 land cover types. These images were extracted from the Tiangong-2

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:46:26 UTC from IEEE Xplore. Restrictions apply.

TABLE II
SUPERVISED COMPARISON RESULTS ON NWPU-RESISC45 BETWEEN SPECIALIST MODELS AND OUR EARTHGPT

Method	Publication Year	Top-1 Acc
Specialist Models		
SeCo [37]	ICCV 2021	92.71
MGSNet [39]	TGRS 2023	**94.57**
CSDS [38]	JSTARS 2021	93.59
T-CNN [40]	TGRS 2022	93.05
PSGAN [41]	TGRS 2022	88.47
MLLM		
EarthGPT(Ours)		93.84

TABLE III
ZERO-SHOT COMPARISON RESULTS BETWEEN EARTHGPT AND OTHER MLLMS ON CLRS AND NASC-TG2

Method	Publication Year	CLRS	NaSC-TG2
Qwen-VL-Chat [98]	Arxiv 2023	51.76	36.09
LLaVa-1.5 [6]	NeurIPS 2023	55.86	40.67
Sphinx [78]	Arxiv 2023	60.72	49.76
EarthGPT(Ours)		**77.37**	**74.72**

satellite and were released by the Center for Space Applications and Engineering of the Chinese Academy of Sciences in 2021. Each image in this dataset has a size of $128\times128\times3$, and the pixel resolution is 100 m. For the evaluation of supervised classification and zero-shot classification, Top-1 accuracy is reported.

In the classification evaluation, similar to the training phase, we present all categories from the datasets as reference categories. The model is instructed to classify images using only one word or phrase. First, we report the accuracy for EarthGPT on the test sets of NWPU-RESISC45 in Table II, comparing its performance with some SOTA specialist models. It is observed that EarthGPT outperforms SOTA specialist models such as CSDS, and T-CNN with improvements of 0.25% and 0.79%, achieving comparable performance with MGSNet. In the zero-shot classification evaluation, we compare EarthGPT with other MLLMs such as QwenVL-Chat, LLaVa-1.5, and Sphinx in Table III. It is apparent that EarthGPT shows a notable improvement in the zero-shot evaluation compared to other MLLMs. Specifically, EarthGPT outperforms Qwen-VL-Chat, LLavaV1.5, and Sphinx in Top-1 accuracy on the CLRS dataset, achieving 25.61%, 21.51%, and 16.65%, respectively. In addition, EarthGPT brings improvements of 38.63%, 34.05%, and 24.96% on the NaSC-TG2 dataset compared with Qwen-VL-Chat, LLavaV1.5, and Sphinx, respectively. This suggests that the integration of domain knowledge from RS is highly important for generalizing to unknown classification scenarios.

C. Image Captioning

For the evaluation of image captioning capability, the test set of the NWPU-Caption dataset is utilized to assess and compare EarthGPT with specialist models in a supervised setting. The NWPU-Caption dataset is created by Northwestern Polytechnical University, incorporating 31 500 aerial RS images and 157 500 sentences for RS image caption. Following the setting of MLCA-Net, we employ BLEU1, BLEU2, BLEU3, BLEU4, METEOR, ROUGEL, and CIDErD as evaluation metrics.

From Table IV, it can be clearly seen that on the NWPU-Captions dataset, compared to other SOTA methods, EarthGPT improves 10.1%, 13.8%, 17.5%, 17.7%, 10.5%, 18.2%, 66.2%, and 3.7% in terms of BLEU1, BLEU2, BLEU3, BLEU4, METEOR, ROUGE-L, and CIDEr-D, respectively. The experimental results in Table IV clearly illustrate that EarthGPT, with its rich and diverse training corpus and more flexible interaction pattern, shows better performance and prospects in the supervised image captioning field compared to previous specialist models.

D. Visual Question Answering

The evaluation of VQA is also divided into supervised and zero-shot assessments. For the supervised VQA setting, we adopt the test set of CRSVQA including 1000 image-question-answer pairs. Following the setting of MQVQA, 10% of image-question-answer pairs are adopted as the test set, and the overall accuracy (OA) is reported. For the zero-shot VQA setting, RSVQA-HR is utilized to evaluate the performance of EarthGPT and other MLLMs. The RSVQA-HR dataset was created from high-resolution orthophoto (HRO) datasets obtained from USGS. It consists of 10 659 images, with 1 066 316 question-answer pairs. RSVQA-HR dataset is divided into training, validation, test set 1, and test set 2. In the evaluation of zero-shot VQA, test set 2 is adopted for evaluating the model's robustness to diverse locations. In addition, following the setting of RSGPT and GeoChat, object counting problems are not included in the computation of metrics for RSVQA-HR. For the RSVQA-HR dataset, the accuracy of different question categories and OA has been reported.

From Table V, it is observed that EarthGPT significantly outperforms other specialist models in terms of OA on CRSVQA datasets. Specifically, EarthGPT surpasses MQVQA and SAN with improvements of 11.09% and 20.83%, respectively. Such significant performance improvement fully demonstrates that, compared with traditional specialist models that treat VQA tasks as classification tasks, generative large models like EarthGPT have stronger capabilities in understanding and responding to images and questions. In zero-shot evaluation, EarthGPT achieves an average accuracy of 72.06% on RSVQA-HR above or commensurate with other MLLMs, as shown in Table VI. Compared with Qwen-VL-Chat, LLava-1.5, minigptV2, and Sphinx, EarthGPT brings the improvement of 3.66%, 9.00%, 25.60%, and 2.27%, respectively. In conclusion, EarthGPT demonstrates exceptional capabilities in answering questions within entirely new and unfamiliar RS contexts, and a profound ability to comprehend the true significance of queries presented in visual data and text instructions.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:46:26 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
SUPERVISED COMPARISON RESULTS ON NWPU CAPTION BETWEEN SPECIALIST MODELS AND OUR MLLM EARTHGPT

Method	Publication Year	BLEU1	BLEU2	BLEU3	BLEU4	METEOR	ROUGE-L	CIDEr(0-5)	SPICE
Specialist Model									
CSMLF [42]	GRSL 2019	77.0	64.9	53.2	47.1	32.0	57.8	106.5	26.5
Qu, *et.al* [71]	CITS 2016	72.5	60.3	51.8	45.5	33.6	59.1	117.9	27.6
Attention(soft) [72]	TGRS 2017	73.1	60.9	52.5	46.2	33.9	59.9	113.6	28.5
Attention(hard) [72]	TGRS 2017	73.3	61.0	52.7	46.4	34.0	60.0	110.3	28.4
FC-Att [43]	TGRS 2019	73.6	61.5	53.2	46.9	33.8	60.0	123.1	28.3
SM-Att [43]	TGRS 2019	73.9	61.7	53.2	46.8	33.0	59.3	123.6	27.6
MLCA-Net [15]	TGRS 2022	74.5	62.4	54.1	47.8	33.7	60.1	126.4	28.5
MLLM									
EarthGPT(Ours)		**87.1**	**78.7**	**71.6**	**65.5**	**44.5**	**78.2**	**192.6**	**32.2**

TABLE V
SUPERVISED COMPARISON RESULTS ON CRSVQA BETWEEN SPECIALIST MODELS AND OUR EARTHGPT

Method	Publication Year	OA
Specialist Model		
Qonly [44]	CVPR 2019	23.49
RSVQA [45]	TGRS 2020	58.96
RSVQA(GRU) [45]	TGRS 2020	59.41
SAN [46]	CVPR 2016	61.17
MQVQA [16]	TGRS 2023	70.91
MLLM		
EarthGPT(Ours)		**82.00**

TABLE VI
ZERO-SHOT COMPARISON RESULTS BETWEEN EARTHGPT AND OTHER MLLMS FOR PRESENCE, COMPARISON, AND OA

Method	Publication Year	Presence	Comparison	OA
Qwen-VL-Chat [97]	Arxiv 2023	**69.83**	67.29	68.40
LLaVa-1.5 [6]	NeurIPS 2023	66.44	60.41	63.06
MiniGPTV2 [76]	Arxiv 2023	40.79	50.91	46.46
Sphinx [77]	Arxiv 2023	64.28	74.75	69.79
GeoChat [11]	CVPR 2024	58.45	**83.19**	**72.30**
EarthGPT(Ours)		62.77	79.53	72.06

E. Visual Grounding

To evaluate the visual grounding performance, we use the test set of DIOR-RSVG containing 758 grounding questions. DIOR-RSVG dataset comprises 38 320 language expressions across 17 402 RS images, uniquely characterizing individual objects within 20 diverse categories. Evaluation metrics include Pr@0.5, Pr@0.6, Pr@0.7, Pr@0.8, Pr@0.9 mean IoU(mIoU), and cum IoU(cIoU). Specialist models ZSGNet, FAOA, ReSc, LBYL-Net, TranVG, VLTVG, and MGVLF are adopted to compare with EarthGPT.

Table VII shows the performance of EarthGPT and other specialist models on the DIOR-RSVG test set. It can be seen that EarthGPT brings significant performance improvements of 0.11%, 1.19%, 1.30%, and 3.13% on Pr@0.8, Pr@0.9, mIoU, and cIoU, respectively. As the precision measurement threshold becomes larger, the performance of EarthGPT increases significantly, compared to other specialist models. This phenomenon indicates that the detection boxes predicted by EarthGPT for visual grounding are more precise. In addition, the noticeable improvements of mIoU and cIoU demonstrate that the overall prediction accuracy of EarthGPT is higher. It is evident that EarthGPT has exceptional spatial location and region-level image comprehension ability.

F. Object Detection

To thoroughly assess the potential and generalization capabilities of the innovative language-guided paradigm of EarthGPT in object detection, we adopt a zero-shot setting for comparison with other MLLMs and specialist models. Given that other MLLMs are primarily designed for HBB detection and struggle with OBB detection, we utilize the HBB format of the MAR20 dataset [20] to evaluate the detection performance, employing AP@40 and AP@50 as metrics. The MAR20 dataset is a large-scale RS aircraft target recognition dataset. It comprises 3842 images with 22 341 instances, and each target instance is annotated with both HBBs and OBBs. Comparing methods include MLLMs Qwen-VL-Chat, Sphinx, Lenna, and open-set object detection models GroundingDINO and mm-GroundingDINO. To address the challenge of MLLMs not predicting confidence scores, we employ clip-score as a confidence logit for noise filtering. Remoteclip [104] weights are adopted to compute the clipscore. For the evaluation of OBB detection, we adopt the OBB format of MAR20 to compare with specialist models trained on DOTA, such as S2A-Net, CFA, Oriented RepPoints, Oriented R-CNN, Sasm, and R3Det.

From Table VIII, on the test set of MAR20, EarthGPT exhibits notable enhancements of AP@40, AP@50 compared to other MLLMs and open-set detection models, underscoring its robust generalization capability under unseen scenarios.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:46:26 UTC from IEEE Xplore. Restrictions apply.

TABLE VII
SUPERVISED COMPARISON RESULTS ON DIOR-RSVG BETWEEN SPECIALIST MODELS AND OUR EARTHGPT

Method	Publication Year	Pr@0.5	Pr@0.6	Pr@0.7	Pr@0.8	Pr@0.9	mIoU	cIoU
Specialist Model								
ZSGNet [47]	ICCV 2019	51.67	48.13	42.30	32.41	10.15	44.12	51.65
FAOA [48]	ICCV 2019	70.86	67.37	62.04	53.19	36.44	62.86	67.28
ReSC [98]	ECCV 2020	72.71	68.92	63.01	53.70	33.37	64.24	68.10
LBYL-Net [49]	CVPR 2021	73.78	69.22	65.56	47.89	15.69	65.92	76.37
TransVG [50]	ICCV 2021	72.41	67.38	60.05	49.10	27.84	63.56	76.27
VLTVG [99]	CVPR 2021	75.79	72.22	66.33	55.17	33.11	66.32	77.85
MGVLF [17]	TGRS 2023	**76.78**	**72.68**	**66.74**	56.42	35.07	68.04	78.41
MLLM								
EarthGPT(Ours)		76.65	71.93	66.52	**56.53**	**37.63**	**69.34**	**81.54**

TABLE VIII
ZERO-SHOT COMPARISON RESULTS FOR HBB DETECTION ON MAR20 BETWEEN OTHER OPEN-SET MODELS AND OUR EARTHGPT

Method	Publication Year	AP@40	AP@50
Open-set Model			
GroundingDINO [100]	Arxiv 2023	88.55	88.21
mm-GroundingDINO [101]	Arxiv 2024	88.37	88.14
MLLM			
Lenna [102]	Arxiv 2023	72.69	72.11
Qwen-VL-Chat [97]	Arxiv 2023	72.56	40.04
Sphinx [77]	Arxiv 2023	81.03	80.48
EarthGPT(Ours)		**90.47**	**90.11**

TABLE IX
ZERO-SHOT COMPARISON RESULTS FOR OBB DETECTION ON MAR20 BETWEEN OTHER METHODS AND OUR EARTHGPT

Method	Publication Year	AP@40	AP@50
Specialist Model			
S2A-Net [51]	TGRS 2021	90.71	90.23
CFA [52]	CVPR 2021	90.66	90.28
Oriented RepPoints [103]	CVPR 2022	90.68	**90.56**
Oriented R-CNN [53]	ICCV 2021	**90.70**	90.54
Sasm [55]	AAAI 2022	90.19	89.83
R3Det [54]	AAAI 2021	90.61	90.11
MLLM			
EarthGPT(Ours)		90.53	87.86

Specifically, EarthGPT brings improvements of 1.92% and 1.90% in terms of AP@40 and AP@50 compared with the open-set model GroundingDINO. For OBB detection, Table IX demonstrates that EarthGPT achieves competitive performance with specialist models trained on DOTA, showcasing the powerful potential of the language-based generative detection paradigm.

G. Ablation Analysis

1) Two-Step Training: This section aims to study the necessity and effectiveness of two-step training designed to realize domain transfer learning. The cross-modal mutual comprehension and unified instruction tuning steps are abbreviated as "Nature Pretraining" and "RS Fine-tuning," respectively. It is clearly shown from Table X that by solely leveraging common nature datasets for pretraining, EarthGPT achieves Top-1 accuracy of 38.21% on the NaSc-TG2 dataset. When directly using MMRS-1M for fine-tuning, EarthGPT achieves a Top-1 accuracy of 67.35%. When integrating nature pretraining and RS fine-tuning, the Top-1 accuracy increases by 7.37%. For other tasks such as VQA, RSVG, and HBB detection, the two-step training strategy brings EarthGPT an increase of 3.82% (OA), 4.06% (mIoU), and 3.74% (AP@50) on RSVQA-HR, DIOR-RSVG, and MAR20 datasets, respectively. The experiment results fully demonstrate the importance of the two-step training strategy for EarthGPT.

2) Visual Encoders: In order to study the influence of different visual encoders on the results, we conduct the ablation experiment, and the results are shown in Table XI. It is evident that mixed encoders bring the best performance in downstream tasks. Specifically, compared to solely using CNN or ViT visual backbone, our designed mixed visual encoders (CLIP ConvNeXt-L and DINOv2 ViT-L/14) improve performance in classification, VQA, RSVG, and HBB detection tasks by 3.35%, 1.43%, 2.43%, and 2.72%, respectively. Moreover, compared to the combination of CLIP RN50X16 and DINOv2 ViT-L/14, our designed mixed visual encoders (CLIP ConvNeXt-L and DINOv2 ViT-L/14) improves performance in classification, VQA, RSVG, and HBB detection by 2.37%, 0.91%, 1.85%, and 1.87%, respectively.

3) Language Decoders: To study the influence of different language decoders on the model performance, we set different LLMs for the ablation experiment as shown in Table XII. Note that the performances of EarthGPT are closing when we use Mixtral-7B and Llama 2-7B as the language decoder. More importantly, when we scale Llama 2-7B up to Llama 2-13B, EarthGPT shows performance improvements in classification, VQA, RSVG, and HBB detection by 4.15%, 2.62%, 4.59%, and 1.19%, respectively.

H. Computation Analysis and Comparisons

In this section, we conduct a computational analysis between EarthGPT and other MLLMs, specialist models.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:46:26 UTC from IEEE Xplore. Restrictions apply.

TABLE X
ABLATION RESULTS ABOUT DESIGNS OF TWO-STEP TRAINING FOR RS

EarthGPT		Classification	VQA	RSVG	HBB detection
Nature Pre-training	RS Fine-tuning	NaSC-TG2 (Top-1 Acc)	RSVQA-HR(OA)	DIOR-RSVG(mIoU)	MAR20(AP@50)
✓	×	38.21	-	-	-
×	✓	67.35	68.24	65.28	86.37
✓	✓	**74.72**	**72.06**	**69.34**	**90.11**

TABLE XI
ABLATION RESULTS ABOUT THE DESIGN OF THE VISUAL ENCODERS

CNN Model		ViT Model	Classification	VQA	RSVG	HBB Detection
CLIP ConvNeXt-L	CLIP RN50×16	DINOv2 ViT-L/14	NaSC-TG2(Top-1 Acc)	RSVQA-HR(OA)	DIOR-RSVG(mIoU)	MAR20(Ap@50)
✓	×	×	70.26	70.63	66.91	86.59
×	✓	×	71.37	69.49	65.03	87.39
×	×	✓	68.56	69.32	64.57	86.46
×	✓	✓	72.35	71.15	67.49	88.24
✓	×	✓	**74.72**	**72.06**	**69.34**	**90.11**

TABLE XII
ABLATION RESULTS ABOUT THE DESIGN OF LANGUAGE DECODER

Language Decoder	Classification	VQA	RSVG	HBB Detection
	NaSC-TG2(Top-1 Acc)	RSVQA-HR(OA)	DIOR-RSVG(mIoU)	MAR20(AP@50)
Mixtral-7B	69.85	69.27	65.72	87.24
Llama 2-7B	70.57	69.44	64.75	88.92
Llama 2-13B	**74.72**	**72.06**	**69.34**	**90.11**

Specifically, we compare the inference time (s) and GPU memory (MiB) usage under the zero-shot setting in two fundamental visual tasks, including image classification and OBB detection. Note that for MLLMs, the GPU memory consumption is primarily driven by storing model weights. As shown in the top of Table XIII, for image classification on the NaSc-TG2 dataset, EarthGPT has a larger GPU memory consumption and longer inference time compared to Qwen-VL-Chat and LLaVa-1.5 and is lower than Sphinx. Moreover, since the MLLMs are language generation paradigms, their inference time increases with the number of words generated, especially in detection tasks where the inference time rises with the increase in the number of detection objects. For OBB detection, most existing MLLMs are incapable of OBB detection due to lacking training data, therefore we choose the specialist models as a contrast. As shown in the bottom part of Table XIII, on the MAR20 dataset, EarthGPT detects 1, 5, and 10 objects averagely need 0.72, 1.60, and 2.01 s, respectively. Due to the model parameters being much larger than those of specialist models, the GPU memory consumption is also much higher than that of specialist models.

TABLE XIII
COMPUTATIONAL ANALYSIS FOR OUR EARTHGPT AND OTHER METHODS

	Method	Inference time (s)	GPU memory (MiB)
Classification	***MLLMs***		
	Qwen-VL-Chat	0.22	38145
	LLava-1.5	0.17	15461
	Sphinx	0.82	41853
	EarthGPT(Ours)	0.71	41776
OBB Detection	***Specialist models***		
	S2A-Net	0.69	2329
	CFA	0.64	2327
	Oriented RepPoints	0.62	2325
	Oriented R-CNN	0.07	2809
	Sasm	0.09	2327
	R3Det	0.13	2177
	MLLMs		
	EarthGPT(Ours) 1 object	0.72	41776
	EarthGPT(Ours) 5 objects	1.60	41776
	EarthGPT(Ours) 10 objects	2.01	41776

VI. VISUALIZATION

In this section, we present the qualitative experimental result of EarthGPT to demonstrate proficiency in multiturn dialog and diverse RS tasks under multisensor visual modalities. EarthGPT showcases a remarkable capacity for image-level and region-level visual perception, and adeptly interpreting RS complex visual data. Notably, EarthGPT also shows advanced performance in chain-of-thought reasoning, fostering the emergence of cross-task visual comprehension.

A. Multiturn Multitask Dialog

Fig. 4 provides a detailed depiction of EarthGPT's capabilities in conducting multiturn dialogs. Diverging from traditional specialist models limited to specific tasks, EarthGPT leverages language as a medium to achieve diverse abilities in multiturn dialog, including classification, image captioning, VQA, object detection, region-level captioning, and visual grounding.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:46:26 UTC from IEEE Xplore. Restrictions apply.

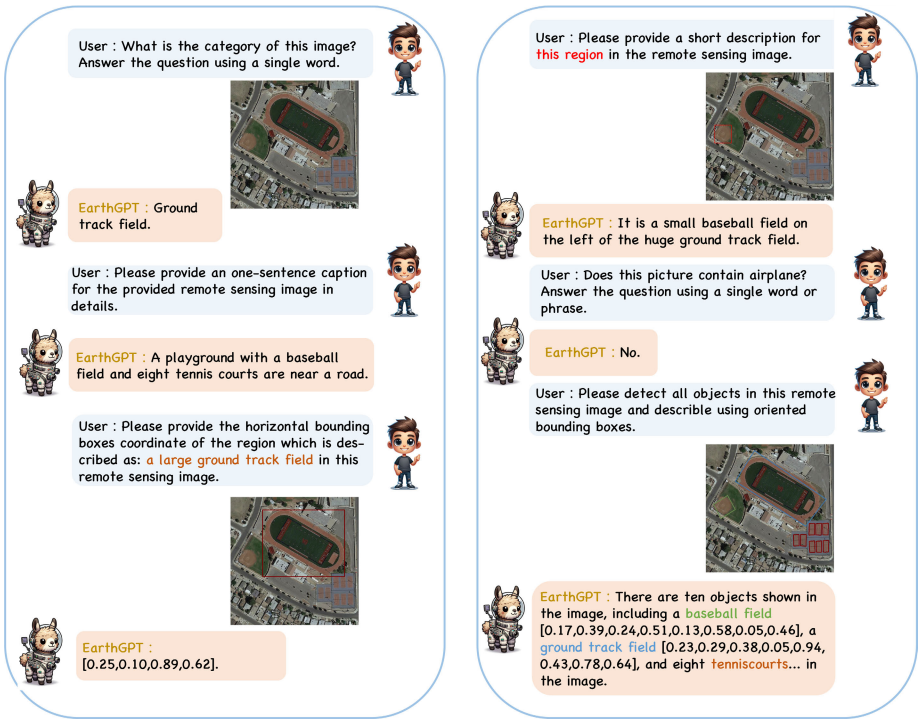


Fig. 4. Examples of EarthGPT for multiturn various RS tasks dialog.

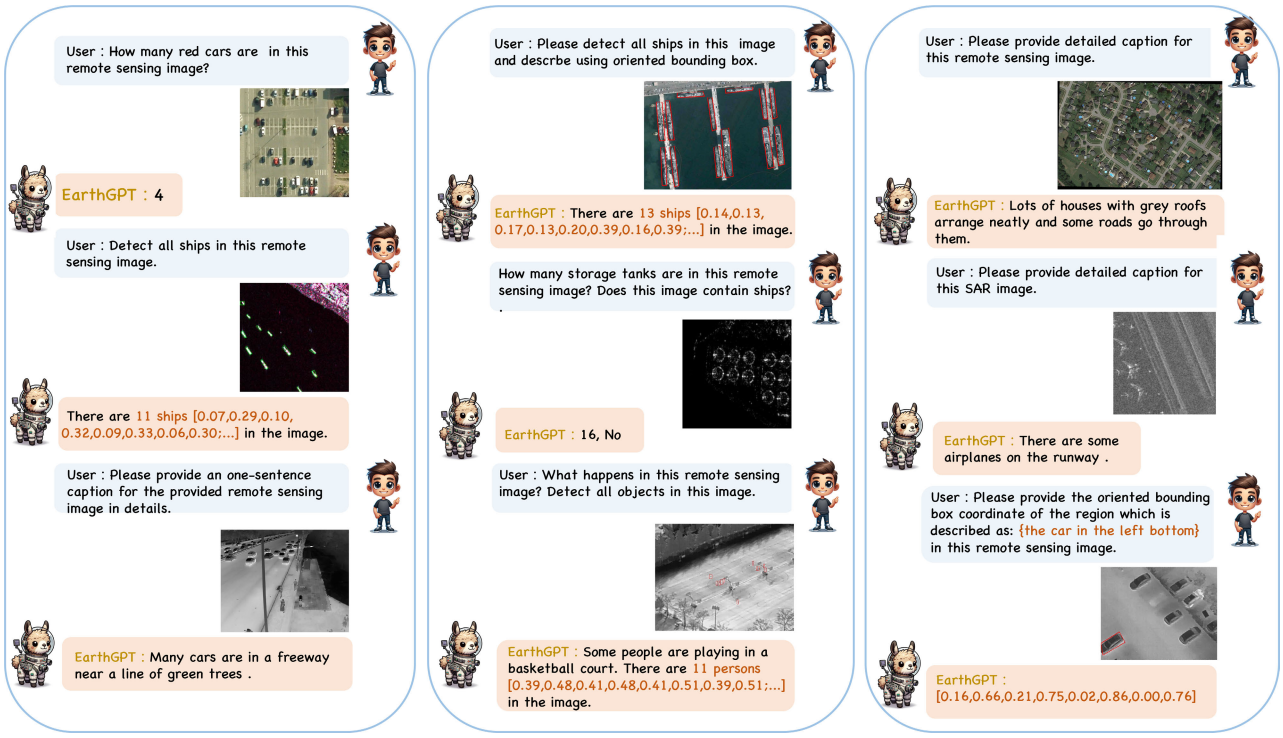


Fig. 5. Examples of EarthGPT for different visual modalities inference ability.

In Fig. 4, we demonstrate EarthGPT's proficiency in accurately identifying the scene as a ground track field, along with a detailed spatial layout description. EarthGPT localizes and describes specific areas precisely, such as the ground track field and the nearby small baseball field. Compared to the previous specialist models, EarthGPT enhances readability and comprehension, making it more user-friendly and accessible.

B. Multi Visual Modality Inference

Previous specialist models in the RS domain mainly concentrate on single visual modality and single task. Thanks to our MMRS-1M's rich multisensor information, laying the foundation for training the versatile EarthGPT. EarthGPT has robust generalization capabilities across optical, SAR, and infrared visual modalities. Fig. 5 shows EarthGPT's

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:46:26 UTC from IEEE Xplore. Restrictions apply.

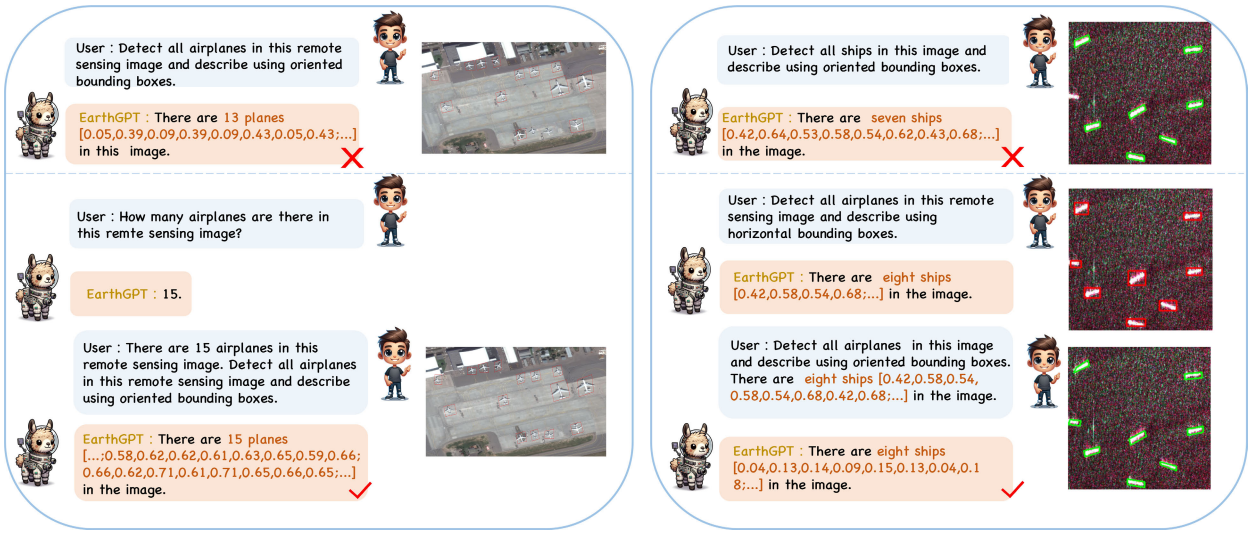


Fig. 6. Examples of the chain-of-thought prompting for EarthGPT to perform visual reasoning.

remarkable performance across a variety of tasks in different visual modalities.

Although the MMRS-1M dataset involves limited SAR and infrared data compared to the optical data. Note that EarthGPT overcomes the constraints by using language to integrate different visual modalities. As shown in Fig. 5, in optical imagery, EarthGPT can accurately enumerate the number of red cars in the scene, identify all ships on water surfaces, and provide detailed descriptions of dense residential. In SAR imagery, EarthGPT accurately counts storage tanks, evaluates ship presence, and concisely describes the airport. In infrared imagery, EarthGPT precisely locates cars in certain areas, identifies persons on a basketball court, and accurately interprets their activities. Those instances illustrate EarthGPT's exceptional knowledge transfer and generalization abilities.

C. Chain-of-Thought Prompting for Visual Reasoning

We discover that EarthGPT can employ chain-of-thought prompting to enhance visual understanding and reasoning accuracy. For instance, EarthGPT can enhance the OBB detection completeness through the result of object counting as the prompt. As shown in Fig. 6, in the airport detection scene, EarthGPT initially misses two planes, when promptly by the question "How many planes are there in the airport?" after discovering there are 15 planes in total, EarthGPT intelligently uses this newfound knowledge to reevaluate the aircraft detection task, ultimately successfully identifying all the planes. In addition, EarthGPT is adept at enhancing the accuracy of OBB detection by harnessing HBB detection results. In the scenario involving SAR ships, EarthGPT effectively detects all ships using HBB but misses ships using OBB. To improve the OBB results, EarthGPT utilizes the HBB detection results as a valuable hint to rectify the omissions and deliver more precise OBB results. In conclusion, the chain-of-thought prompting technique facilitates intelligently incorporating insights from related tasks to elevate the performance in various visual interpretation and reasoning tasks.

D. Comparison Between EarthGPT and Other MLLMs

In this section, we compare the performance of EarthGPT with other MLLMs when facing challenging samples in the RS domain. Specifically, we choose image classification and object detection scenarios using the zero-shot setting to demonstrate EarthGPT's superior performance. As shown in the left part in Fig. 7, when facing an image of a storage tank, due to the small scale of the storage tank and the interference of other surrounding objects, MLLMs like LLava-1.5 [6], Sphinx [77], and Qwen-VL-Chat [97] incorrectly categorize the storage tank as a stadium or residential area. In contrast, EarthGPT has better visual perception for small-scale targets and can correctly identify the category. In the detection scenario (the right part in Fig. 7), when facing densely packed small-scale airplanes, MLLMs like Lenna [102], Qwen-VL-Chat, and Sphinx all struggle to recognize all the airplane targets and also provide low-precision detection boxes. Note that EarthGPT not only detects all airplane objects but also uses more accurate rotating boxes to pinpoint their locations.

E. Failure Cases of EarthGPT

The previous visualization examples demonstrate that our proposed EarthGPT possesses strong multiturn multitask reasoning capabilities. Moreover, compared to other MLLMs, EarthGPT has superior visual understanding capabilities. However, there still exists failure OBB detection cases in certain complex scenarios. As illustrated in Fig. 8, EarthGPT has limitations in its detection accuracy, for example, there are missed detections in the first image. In addition, in the second ship image, some detection boxes mistakenly include multiple ships. Similarly, other specialist models also suffer from those OBB detection inaccuracies. For example, S2A-Net [51], Sasm [55], and R3Det [54] generate false alarms and missed detection. Note that Oriented R-CNN [53] obtains a better performance in OBB detection accuracy. In summary, the proposed EarthGPT represents a new visual-language paradigm for RS image interpretation and is adept at handling

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:46:26 UTC from IEEE Xplore. Restrictions apply.

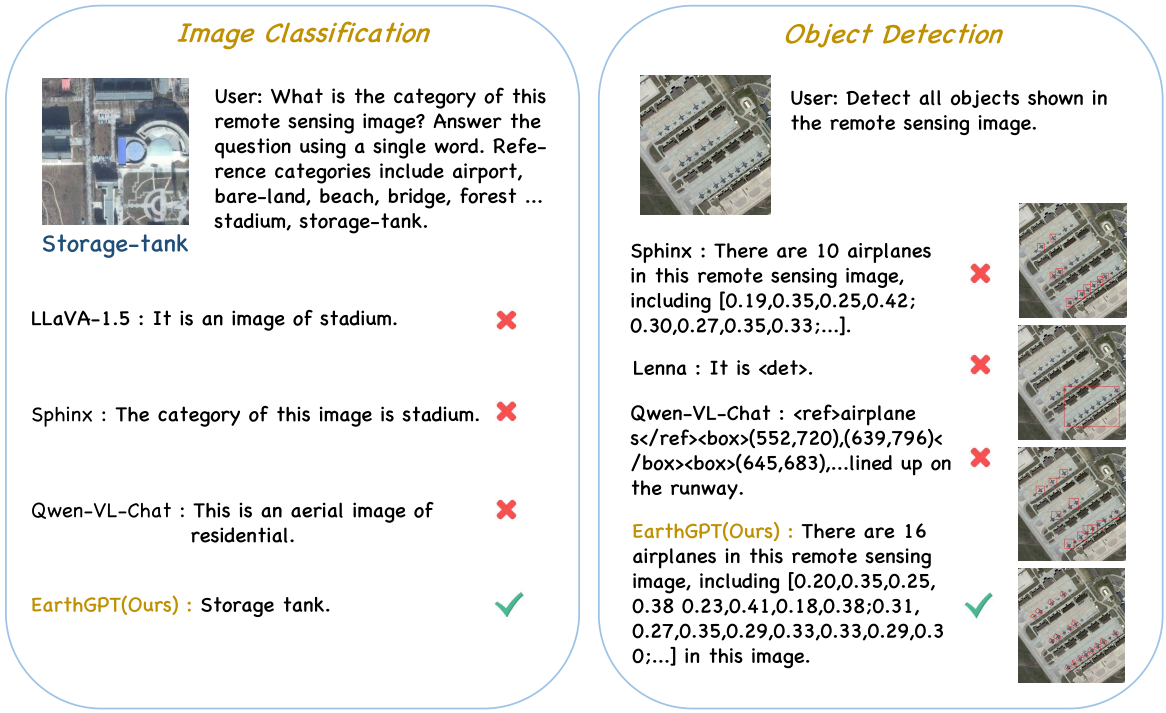


Fig. 7. Examples for performance comparison between EarthGPT and other MLLMs.

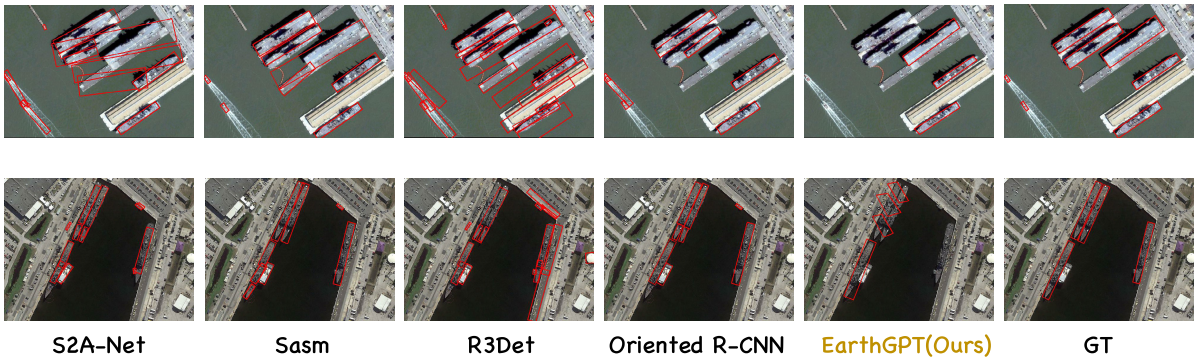


Fig. 8. Failure cases of EarthGPT.

various tasks. However, there are still existing failure cases on complex issues that need improvement. In the future, we will further explore enhancing EarthGPT's OBB detection ability in challenging scenarios.

VII. Conclusion

In this article, a versatile MLLM EarthGPT, which unifies a wide range of RS tasks and various multisensor images, has been proposed for universal RS image comprehension. EarthGPT integrates coarse-scale and fine-scale visual perception information, bridging the gaps in cross-modal mutual comprehension and vision reasoning. More importantly, EarthGPT is capable of achieving multisensor image interpretation and RS downstream tasks including scene classification, image captioning, region-level captioning, VQA, visual grounding, and object detection. Furthermore, the MMRS-1M dataset, a multisensor multimodal RS instruction-following dataset comprising more than 1M image–text pairs, has been constructed. The MMRS-1M dataset tackles the limitation of MLLMs on RS expert knowledge and encourages the growth of MLLMs tailored specifically for applications in the RS domain. Extensive experiments have demonstrated that EarthGPT surpasses the numerous existing specialist models and MLLMs in various RS visual interpretation tasks, and provides an open-set reasoning capability suitable for multiple RS downstream tasks, both in supervised and zero-shot settings. In the future, we will focus on further enhancing the object detection performance of EarthGPT. Furthermore, we plan to incorporate more modalities and a broader range of tasks into EarthGPT to enhance all-purpose capabilities.

References

[1] L. Ma, Y. Liu, X. Zhang, Y. Ye, G. Yin, and B. A. Johnson, "Deep learning in remote sensing applications: A meta-analysis and review," *ISPRS J. Photogramm. Remote Sens.*, vol. 152, pp. 166–177, Jun. 2019.

[2] X. X. Zhu et al., "Deep learning in remote sensing: A comprehensive review and list of resources," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 4, pp. 8–36, Dec. 2017.

[3] T. Zhang et al., "Consecutive pre-training: A knowledge transfer learning strategy with relevant unlabeled data for remote sensing domain," *Remote Sens.*, vol. 14, no. 22, p. 5675, Nov. 2022.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:46:26 UTC from IEEE Xplore. Restrictions apply.

[4] J. Chen, H. Guo, K. Yi, B. Li, and M. Elhoseiny, "VisualGPT: Data-efficient adaptation of pretrained language models for image captioning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 18030–18040.

[5] J. Li, D. Li, S. Savarese, and S. Hoi, "BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models," 2023, *arXiv:2301.12597*.

[6] H. Liu, C. Li, Q. Wu, and Y. Jae Lee, "Visual instruction tuning," 2023, *arXiv:2304.08485*.

[7] J.-B. Alayrac et al., "Flamingo: A visual language model for few-shot learning," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 35, 2022, pp. 23716–23736.

[8] H. Zhao et al., "MMICL: Empowering vision-language model with multi-modal in-context learning," 2023, *arXiv:2309.07915*.

[9] D. Zhu, J. Chen, X. Shen, X. Li, and M. Elhoseiny, "MiniGPT-4: Enhancing vision-language understanding with advanced large language models," 2023, *arXiv:2304.10592*.

[10] Y. Hu, J. Yuan, C. Wen, X. Lu, and X. Li, "RSGPT: A remote sensing vision language model and benchmark," 2023, *arXiv:2307.15266*.

[11] K. Kuckreja, M. S. Danish, M. Naseer, A. Das, S. Khan, and F. Shahbaz Khan, "GeoChat: Grounded large vision-language model for remote sensing," 2023, *arXiv:2311.15826*.

[12] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[13] K. O'Shea and R. Nash, "An introduction to convolutional neural networks," 2015, *arXiv:1511.08458*.

[14] F. Zhuang et al., "A comprehensive survey on transfer learning," *Proc. IEEE*, vol. 109, no. 1, pp. 43–76, Jul. 2020.

[15] Q. Cheng, H. Huang, Y. Xu, Y. Zhou, H. Li, and Z. Wang, "NWPU-captions dataset and MLCA-Net for remote sensing image captioning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5629419.

[16] M. Zhang, F. Chen, and B. Li, "Multistep question-driven visual question answering for remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4704912.

[17] Y. Zhan, Z. Xiong, and Y. Yuan, "RSVG: Exploring data and models for visual grounding on remote sensing data," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5604513.

[18] H. Li et al., "CLRS: Continual learning benchmark for remote sensing image scene classification," *Sensors*, vol. 20, no. 4, p. 1226, Feb. 2020.

[19] Z. Zhou et al., "NaSC-TG2: Natural scene classification with Tiangong-2 remotely sensed imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 3228–3242, 2021.

[20] W. Yu et al., "MAR20: A benchmark for military aircraft recognition in remote sensing images," *Nat. Remote Sens. Bull.*, vol. 27, no. 12, pp. 2688–2696, 2023.

[21] J. Achiam et al., "GPT-4 technical report," 2023, *arXiv:2303.08774*.

[22] H. Touvron et al., "LLaMA: Open and efficient foundation language models," 2023, *arXiv:2302.13971*.

[23] R. Zhang et al., "LLaMA-adapter: Efficient fine-tuning of language models with zero-init attention," 2023, *arXiv:2303.16199*.

[24] H. Touvron et al., "Llama 2: Open foundation and fine-tuned chat models," 2023, *arXiv:2307.09288*.

[25] B. Peng, C. Li, P. He, M. Galley, and J. Gao, "Instruction tuning with GPT-4," 2023, *arXiv:2304.03277*.

[26] S. Huang et al., "Language is not all you need: Aligning perception with language models," 2023, *arXiv:2302.14045*.

[27] P. Gao et al., "LLaMA-adapter v2: Parameter-efficient visual instruction model," 2023, *arXiv:2304.15010*.

[28] Q. Ye et al., "MPLUG-owl: Modularization empowers large language models with multimodality," 2023, *arXiv:2304.14178*.

[29] C. Lyu et al., "Macaw-LLM: Multi-modal language modeling with image, audio, video, and text integration," 2023, *arXiv:2306.09093*.

[30] Z. Tang, Z. Yang, M. Khademi, Y. Liu, C. Zhu, and M. Bansal, "CoDi-2: In-context, interleaved, and interactive any-to-any generation," 2023, *arXiv:2311.18775*.

[31] R. Xu, X. Wang, T. Wang, Y. Chen, J. Pang, and D. Lin, "PointLLM: Empowering large language models to understand point clouds," 2023, *arXiv:2308.16911*.

[32] Z. Guo et al., "Point-bind & point-LLM: Aligning point cloud with multi-modality for 3D understanding, generation, and instruction following," 2023, *arXiv:2309.00615*.

[33] Y. Hong et al., "3D-LLM: Injecting the 3D world into large language models," 2023, *arXiv:2307.12981*.

[34] S. Yang et al., "LiDAR-LLM: Exploring the potential of large language models for 3D LiDAR understanding," 2023, *arXiv:2312.14074*.

[35] M. Ahn et al., "AutoRT: Embodied foundation models for large scale orchestration of robotic agents," 2024, *arXiv:2401.12963*.

[36] W. Dai et al., "InstructBLIP: Towards general-purpose vision-language models with instruction tuning," 2023, *arXiv:2305.06500*.

[37] O. Manas, A. Lacoste, X. Giro-i-Nieto, D. Vazquez, and P. Rodriguez, "Seasonal contrast: Unsupervised pre-training from uncurated remote sensing data," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9414–9423.

[38] X. Wang, L. Yuan, H. Xu, and X. Wen, "CSDS: End-to-end aerial scenes classification with depthwise separable convolution and an attention mechanism," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 10484–10499, 2021.

[39] J. Wang, W. Li, M. Zhang, R. Tao, and J. Chanussot, "Remote-sensing scene classification via multistage self-guided separation network," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5615312.

[40] W. Wang, Y. Chen, and P. Ghamisi, "Transferring CNN with adaptive learning for remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5533918.

[41] G. Cheng, X. Sun, K. Li, L. Guo, and J. Han, "Perturbation-seeking generative adversarial networks: A defense framework for remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–11, 2021.

[42] B. Wang, X. Lu, X. Zheng, and X. Li, "Semantic descriptions of high-resolution remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 8, pp. 1274–1278, Aug. 2019.

[43] X. Zhang, X. Wang, X. Tang, H. Zhou, and C. Li, "Description generation for remote sensing images using attribute attention mechanism," *Remote Sens.*, vol. 11, no. 6, p. 612, Mar. 2019.

[44] K. Marino, M. Rastegari, A. Farhadi, and R. Mottaghi, "OK-VQA: A visual question answering benchmark requiring external knowledge," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3195–3204.

[45] S. Lobry, D. Marcos, J. Murray, and D. Tuia, "RSVQA: Visual question answering for remote sensing data," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 12, pp. 8555–8566, Dec. 2020.

[46] K. Kafle and C. Kanan, "Answer-type prediction for visual question answering," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 4976–4984.

[47] A. Sadhu, K. Chen, and R. Nevatia, "Zero-shot grounding of objects from natural language queries," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 4693–4702.

[48] Z. Yang, B. Gong, L. Wang, W. Huang, D. Yu, and J. Luo, "A fast and accurate one-stage approach to visual grounding," in *Proc. Int. Conf. Comput. Vis.*, Jun. 2019, pp. 4683–4693.

[49] B. Huang, D. Lian, W. Luo, and S. Gao, "Look before you leap: Learning landmark features for one-stage visual grounding," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 16883–16892.

[50] J. Deng, Z. Yang, T. Chen, W. Zhou, and H. Li, "TransVG: End-to-end visual grounding with transformers," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2021, pp. 1769–1779.

[51] J. Han, J. Ding, J. Li, and G.-S. Xia, "Align deep features for oriented object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5602511.

[52] Z. Guo, C. Liu, X. Zhang, J. Jiao, X. Ji, and Q. Ye, "Beyond bounding-box: Convex-hull feature adaptation for oriented and densely packed object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 8792–8801.

[53] X. Xie, G. Cheng, J. Wang, X. Yao, and J. Han, "Oriented R-CNN for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 3520–3529.

[54] X. Yang, J. Yan, Z. Feng, and T. He, "R3Det: Refined single-stage detector with feature refinement for rotating object," in *Proc. AAAI Conf. Artif. Intell.*, vol. 35, 2021, pp. 3163–3171.

[55] L. Hou, K. Lu, J. Xue, and Y. Li, "Shape-adaptive selection and measurement for oriented object detection," in *Proc. AAAI Conf. Artif. Intell.*, 2022, pp. 923–932.

[56] G.-S. Xia et al., "AID: A benchmark data set for performance evaluation of aerial scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3965–3981, Jul. 2017.

[57] P. Helber, B. Bischke, A. Dengel, and D. Borth, "EuroSAT: A novel dataset and deep learning benchmark for land use and land cover classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 7, pp. 2217–2226, Jul. 2019.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:46:26 UTC from IEEE Xplore. Restrictions apply.

[58] G. Cheng, J. Han, and X. Lu, "Remote sensing image scene classification: Benchmark and state of the art," *Proc. IEEE*, vol. 105, no. 10, pp. 1865–1883, Oct. 2017.

[59] Y. Yang and S. Newsam, "Bag-of-visual-words and spatial extensions for land-use classification," in *Proc. 18th SIGSPATIAL Int. Conf. Adv. Geograph. Inf. Syst.*, Nov. 2010, pp. 270–279.

[60] D. Dai and W. Yang, "Satellite image classification via two-layer sparse coding with biased image representation," *IEEE Geosci. Remote Sens. Lett.*, vol. 8, no. 1, pp. 173–176, Jan. 2011.

[61] J. Han, D. Zhang, G. Cheng, L. Guo, and J. Ren, "Object detection in optical remote sensing images based on weakly supervised learning and high-level feature learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 6, pp. 3325–3337, Jun. 2015.

[62] J. Ding et al., "Object detection in aerial images: A large-scale benchmark and challenges," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 11, pp. 7778–7796, Nov. 2022.

[63] X. Sun et al., "FAIR1M: A benchmark dataset for fine-grained object recognition in high-resolution remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 184, pp. 116–130, Feb. 2022.

[64] Y. Zhang, Y. Yuan, Y. Feng, and X. Lu, "Hierarchical and robust convolutional neural network for very high-resolution remote sensing object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5535–5548, Aug. 2019.

[65] K. Li, G. Cheng, S. Bu, and X. You, "Rotation-insensitive and context-augmented object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 4, pp. 2337–2348, Apr. 2017.

[66] S. Xian, W. Zhirui, S. Yuanrui, D. Wenhui, Z. Yue, and F. Kun, "AIR-SARShip-1.0: High-resolution SAR ship detection dataset," *Radars*, vol. 8, no. 6, pp. 852–863, 2019.

[67] S. Wei, X. Zeng, Q. Qu, M. Wang, H. Su, and J. Shi, "HRSID: A high-resolution SAR images dataset for ship detection and instance segmentation," *IEEE Access*, vol. 8, pp. 120234–120254, 2020.

[68] T. Zhang et al., "SAR ship detection dataset (SSDD): Official release and comprehensive data analysis," *Remote Sens.*, vol. 13, no. 18, p. 3690, Sep. 2021.

[69] S. W. Zamir et al., "iSAID: A large-scale dataset for instance segmentation in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops*, Jul. 2019, pp. 28–37.

[70] J. Wang, Z. Zheng, A. Ma, X. Lu, and Y. Zhong, "LoveDA: A remote sensing land-cover dataset for domain adaptive semantic segmentation," in *Proc. Neural Inf. Process. Syst. Track Datasets Benchmarks*, vol. 1, J. Vanschoren and S. Yeung, Eds. Red Hook, NY, USA: Curran Associates, 2021.

[71] B. Qu, X. Li, D. Tao, and X. Lu, "Deep semantic understanding of high resolution remote sensing image," in *Proc. Int. Conf. Comput., Inf. Telecommun. Syst. (CITS)*, Jul. 2016, pp. 1–5.

[72] X. Lu, B. Wang, X. Zheng, and X. Li, "Exploring models and data for remote sensing image caption generation," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 4, pp. 2183–2195, Apr. 2018.

[73] Z. Yuan et al., "Exploring a fine-grained multiscale method for cross-modal remote sensing image retrieval," 2022, *arXiv:2204.09868*.

[74] M. Rahnemoonfar, T. Chowdhury, A. Sarkar, D. Varshney, M. Yari, and R. R. Murphy, "FloodNet: A high resolution aerial imagery dataset for post flood scene understanding," *IEEE Access*, vol. 9, pp. 89644–89654, 2021.

[75] X. Zheng, B. Wang, X. Du, and X. Lu, "Mutual attention inception network for remote sensing visual question answering," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5606514.

[76] J. Chen et al., "MiniGPT-v2: Large language model as a unified interface for vision-language multi-task learning," 2023, *arXiv:2310.09478*.

[77] Z. Lin et al., "SPHINX: The joint mixing of weights, tasks, and visual embeddings for multi-modal large language models," 2023, *arXiv:2311.07575*.

[78] T. Kudo and J. Richardson, "SentencePiece: A simple and language independent subword tokenizer and detokenizer for neural text processing," 2018, *arXiv:1808.06226*.

[79] T. Mikolov, K. Chen, G. Corrado, and J. Dean, "Efficient estimation of word representations in vector space," 2013, *arXiv:1301.3781*.

[80] C. Schuhmann et al., "LAION-400M: Open dataset of CLIP-filtered 400 million image-text pairs," 2021, *arXiv:2111.02114*.

[81] X. Chen et al., "Microsoft COCO captions: Data collection and evaluation server," 2015, *arXiv:1504.00325*.

[82] Q. Zou, L. Ni, T. Zhang, and Q. Wang, "Deep learning based feature selection for remote sensing scene classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 11, pp. 2321–2325, Nov. 2015.

[83] Y. H. Di, Z. G. Jiang, and H. P. Zhang, "A public dataset for fine-grained ship classification in optical remote sensing images," *Remote Sens.*, vol. 13, no. 4, p. 747, 2021.

[84] Y. Di, Z. Jiang, H. Zhang, and G. Meng, "A public dataset for ship classification in remote sensing images," *Proc. SPIE*, vol. 11155, pp. 515–521, Oct. 2019.

[85] Y. Long, Y. Gong, Z. Xiao, and Q. Liu, "Accurate object localization in remote sensing images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 5, pp. 2486–2498, May 2017.

[86] H. Zhu, X. Chen, W. Dai, K. Fu, Q. Ye, and J. Jiao, "Orientation robust object detection in aerial images using deep convolutional neural network," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2015, pp. 3735–3739.

[87] P. Zhu et al., "Detection and tracking meet drones challenge," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 11, pp. 7380–7399, Nov. 2022.

[88] J. Suo, T. Wang, X. Zhang, H. Chen, W. Zhou, and W. Shi, "HIT-UAV: A high-altitude infrared thermal dataset for unmanned aerial vehicle-based object detection," *Sci. Data*, vol. 10, no. 1, p. 227, Apr. 2023.

[89] InfiRay. (2021). *Sea-Shipping*. [Online]. Available: http://openai.iraytek.com/apply/Sea_shipping.html/

[90] InfiRay. (2021). *Infrared-Security*. [Online]. Available: http://openai.iraytek.com/apply/Infrared_security.html/

[91] InfiRay. (2021). *Aerial-Mancar*. [Online]. Available: http://openai.raytrontek.com/apply/Aerial_mancar.html/

[92] InfiRay. (2021). *Double-Light-Vehicle*. [Online]. Available: http://openai.raytrontek.com/apply/Double_light_vehicle.html/

[93] Center for Optics Research and Engineering of Shandong University. (2020). *Oceanic-Ship*. [Online]. Available: http://www.gxzx.sdu.edu.cn/info/1133/2174.htm/

[94] M. Oquab et al., "DINOv2: Learning robust visual features without supervision," 2023, *arXiv:2304.07193*.

[95] G. Ilharco et al., Jul. 2021, "OpenCLIP," *Zenodo*, doi: 10.5281/zenodo.5143772.

[96] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.

[97] J. Bai et al., "Qwen-VL: A versatile vision-language model for understanding, localization, text reading, and beyond," 2023, *arXiv:2308.12966*.

[98] Z. Yang, T. Chen, L. Wang, and J. Luo, "Improving one-stage visual grounding by recursive sub-query construction," in *Proc. 16th Eur. Conf. Comput. Vis.* Glasgow, U.K.: Springer, Aug. 2020, pp. 387–404.

[99] L. Yang, Y. Xu, C. Yuan, W. Liu, B. Li, and W. Hu, "Improving visual grounding with visual-linguistic verification and iterative reasoning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 9489–9498.

[100] S. Liu et al., "Grounding DINO: Marrying DINO with grounded pre-training for open-set object detection," 2023, *arXiv:2303.05499*.

[101] X. Zhao et al., "An open and comprehensive pipeline for unified object grounding and detection," 2024, *arXiv:2401.02361*.

[102] F. Wei, X. Zhang, A. Zhang, B. Zhang, and X. Chu, "Lenna: Language enhanced reasoning detection assistant," 2023, *arXiv:2312.02433*.

[103] W. Li, Y. Chen, K. Hu, and J. Zhu, "Oriented reppoints for aerial object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 1829–1838.

[104] F. Liu et al., "RemoteCLIP: A vision language foundation model for remote sensing," 2023, *arXiv:2306.11029*.



Wei Zhang received the M.E. degree from the Renmin University of China, Beijing, China, in 2020. She is currently pursuing the Ph.D. degree with the Advanced Research Institute of Multidisciplinary Science, Beijing Institute of Technology, Beijing.

Her research interests include multimodal large language models and their applications in remote sensing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:46:26 UTC from IEEE Xplore. Restrictions apply.



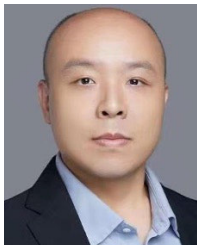
Miaoxin Cai (Graduate Student Member, IEEE) was born in Gansu, China, in 2000. He received the B.S. degree from Northwestern Polytechnical University, Xi'an, China, in 2022. He is currently pursuing the Ph.D. degree in information and communication engineering with the Beijing Institute of Technology, Beijing, China.

His research interests include remote sensing visual language models, remote sensing semi-supervised learning, and remote sensing semantic segmentation.



Tong Zhang (Graduate Student Member, IEEE) was born in Hebei, China, in 1997. He received the B.S. degree from the Hebei University of Technology, Tianjin, China, in 2019. He is currently pursuing the Ph.D. degree in information and communication engineering with the Beijing Institute of Technology, Beijing, China.

His research interests include remote sensing object detection and recognition.



Yin Zhuang (Member, IEEE) was born in Henan, China, in 1990. He received the B.S. degree in electronic engineering from the University of Sussex, Brighton, U.K., in 2013, and the Ph.D. degree from the Beijing Institute of Technology, Beijing, China, in 2018.

From 2018 to 2020, he was a Post-Doctoral Researcher with the School of Electronic Engineering and Computer Sciences, Peking University, Beijing. Since January 2021, he has been a Tenure-Track Assistant Professor with the School of Information and Communication Engineering, Beijing Institute of Technology. His research interests include remote sensing object detection and recognition.



Xuerui Mao received the B.E. and M.E. degrees from the Beijing Institute of Technology, Beijing, China, in 2004 and 2006, respectively, and the Ph.D. degree from Imperial College London, London, U.K., in 2010.

In 2011 and 2012, he was a Research Fellow at Monash University, Melbourne, VIC, Australia, and École Polytechnique, Palaiseau, France, respectively. He was a Lecturer at Durham University, Durham, U.K., from 2012 to 2016. Since 2016, he has been serving as an Assistant Professor with the University of Nottingham, Nottingham, U.K. Since 2021, he has been with the Beijing Institute of Technology, where he is currently a Professor with the Advanced Research Institute of Multidisciplinary Science. His research interests include high-dimensional data modal identification and decomposition, multiprecision data fusion, and multimodal large language models and their applications in remote sensing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:46:26 UTC from IEEE Xplore. Restrictions apply.